Conditional Assembly of Gravity, Matter, and Abelian Gauge Terminals from One Ordered Tau Descent

Tau Core Foundation Paper VII-A

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Abstract

We ask whether the previously constructed Tau gravity, temporal, quantum, matter and Abelian gauge terminals can descend from one ordered physical source ledger rather than from independently fitted sector models. On a stabilized morphological body we construct one once-counted post-body effective functional whose effect, metric, gauge and clock-character derivatives generate probabilities, stress, current and formal clock response. Diffeomorphism and $U(1)$ gauge invariance yield the conditional Einstein–Maxwell–matter equations, Ward/Bianchi identities and their standard weak-field leaf. A disjoint occupation ledger prevents duplicate classical/quantum stress, while exact countermodels show that standard-limit agreement does not establish common Tau provenance.

On an occupied two-mode restriction, ROOT one-particle coherence and transverse-traceless stress arise from one preparation without an additional cross-terminal gain. Differential recovery is reduced to a source-rank problem: after de-duplication, the ROOT increment over the common P1 row space is fixed by one source-owned mixed Gram. The paper therefore gives a conditional common-source assembly and a precise non-identifiability boundary, not a first-principles derivation of Einstein–Hilbert or Maxwell action shapes, not quantized geometry and not unrestricted physical parent realization. The rank-seven P3 packet and scalar-15/RG specialization are now owned by Technical Papers VII-B and VII-C respectively. The inherited signed rank-four clock/body section likewise does not identify projective-state measure with body four-volume; terminal assembly cannot supply the missing connector. We also prove a general observer-access factorization theorem: equal complete access kernels define one parent quotient and unique image transports, whose terminal decomposition may be observer-dependent. This is information-level mixing only. Contextual quantum structure cannot be fully identified with a classical noncontextual gravity terminal by access change alone.

Inherited morphology terminology. The "stabilized morphological body" consumed below is, in Paper I's precise language, the complete *stabilized morphological response configuration* $M_\tau^\star$. The base is its pre-readout carrier/support context and the seed is the corresponding base-readable relational/loading pattern. The terminal assembly assumes the atemporally selected body packet; it derives neither literal morphogenesis nor an observer clock from that selection. Under MRC-DEF2, the parent-side base and seed are typed pregeometric relational candidates. They are not ordinary matter or apparatus objects placed upstream of the geometry and matter terminals that this paper assembles.

Inputs from earlier papers. Paper I supplies body-first descent; Paper II typed finite support; Paper III recovery-correctable record transport; Paper IV the parent-realization boundary; Paper V observer time; Paper VI the conditional quantum terminal. Paper V also supplies the common M4 metrological quotient: one measured dimension-bearing anchor selects an SI representative while all dimensionless clock–ruler terminals remain fixed. Technical Paper VII-B owns the later pointed-exterior pair/top closure and its class-relative occupation theorem; the present paper uses only its support consequence and does not re-prove it.
Results owned here. Morphology-conditioned joint-terminal assembly, the finite occupation criterion, one effective source, the disjoint stress ledger, Ward/Bianchi consistency, standard-form classification inside the declared leaf, ROOT–TT common preparation, Abelian directional activation, the exact de-duplicated source-rank obstruction, and the general observer-dependent terminal-mixing theorem and no-go boundary.

Claim boundary. The Einstein–Hilbert and Maxwell action shapes are inputs of the declared standard Lorentzian leaf and are not derived from the narrow Tau parent. The results below are conditional assembly theorems. They do not prove that the unrestricted physical base–seed law realizes or occupies the common packet, quantize the metric, derive non-Abelian gauge occupation or reproduce the Standard-Model spectrum. They also do not construct a full projective/body measure connector or derive absolute length, time or action units from equality of terminal ranks. The inherited common M4 unit frame is defined only up to its positive metrological orbit; this paper neither adds another terminal gain nor predicts the decimal SI representative. Dimensionless gauge normalization remains a separate source-Hessian/spectral question. Temporal actuality is also an upstream enriched source clause, not an extra stress–energy component generated by terminal assembly. EOCC may order an adopted occurrence–action completion, but the equations assembled here do not select literal Parent traversal, and its proto-order current must not be identified with dark matter, dark energy or a continuing 4D energy injection without a separately derived terminal stress map.

Notation and packet map

Symbol or packet	Role
$P1$	Gravity–time–radiation/matter differential packet.
ROOT	Occupied two-mode quantum one-particle descriptor; its normalized shape is ρ_{ROOT} and its occupation magnitude is retained separately.
TT	Transverse–traceless metric/stress readout paired with ROOT.
$P3$	Rank-seven primitive algebra/recovery packet, imported only at the handoff from Technical Paper VII-B.
$J^{\mathrm{lin}}_{\mathrm{phys}}$	Parent-side physical incidence on the source-frozen linear quotient.
K_*	Positive source metric/Hessian on the true-null quotient.

1 Question and reading order

Inherited observer co-descent convention. The observer is not an external endpoint added to the joint terminal ledger. The symbol O first labels a body-side carrier candidate; regular local rank-four descent, stable observer–source quantization and a nonzero record/effect jointly make it operational and co-produce its accessible 4D world. The terminal maps assembled here act on that occupied common support. They do not create an observer, a body or an independent channel layer.

Typed descriptor and ontology-migration boundary

The shared differentiable object used by the local assembly is

$$\Xi_{OS}^{\mathrm{cont}} = \Phi_{OS}[M_\tau], \qquad D_{OS}^{\mathrm{op}} = Q_{OS,\delta}(\Xi_{OS}^{\mathrm{cont}}). \tag{1}$$

The finite-resolution operational record is therefore not identified with the continuous descriptor. Differential ranks, Hessians and coframes below refer to Ξ_{OS}^{cont}; a hard $Q_{OS,\delta}$ is governed by exact partition fibres.

Let $L : X_{OS} \to Y$ be a historical parent descriptor on the same frozen occupied observer–source domain. The quotient universal property gives a unique realized-image map U with

$$L = U \circ \Xi_{OS}^{\mathrm{cont}} \tag{2}$$

if and only if every Ξ_{OS}^{cont} fibre is contained in an L fibre. On a regular linearized patch this is

$$\ker D\Xi_{OS}^{\mathrm{cont}} \subseteq \ker DL. \tag{3}$$

Equality of the two fibre relations gives exact descriptor equivalence and, on equal-rank smooth strata, a typed isomorphism. Equal rank by itself is insufficient: $\Xi(x, y) = x$ and $L(x, y) = x + y$ have equal rank but incompatible fibres. Thus legacy time, distance and joint-terminal formulas survive only as transported descendants through U. This imports an already sourced coframe, connection, inner product and calibration; it derives none of them, and it does not select the physical $Q_{OS,\delta}$, a galactic $q_R(R)$ or Nature occupation.

The regular four-dimensional support is itself a conditional upstream input to this assembly. In a derivation representation one first needs a common body-side family $\mathfrak{d}_\tau$, a source-owned restriction $R_{OS}^{\mathfrak{d}}$ and a locally constant, integrable visible quotient

$$T_{OS,x}^{\mathrm{vis}} = \mathrm{im}\Big(\widetilde{\Phi}_{OS,x} \circ R_{OS}^{\mathfrak{d}}\Big), \qquad \dim T_{OS,x}^{\mathrm{vis}} = 4.$$

This rank condition does not prove Lorentz signature. The source-owned cone/order must separately admit a time-orientable Lorentz-conformal realization (open gate G4a); standard causal-reconstruction uniqueness then applies only conditionally (G4b). The result is generally a local coframe atlas. A global tetrad requires a trivial frame bundle, and fermionic leaves require a compatible spin structure. Finally a source-owned $U_{\mathrm{DA}\leftrightarrow\mathrm{ROOT}}$ must identify this DA carrier with the existing ROOT/M4TR carrier while preserving nulls, order and conformal geometry. Paper VII-A assumes this handoff; it does not derive or fit it.

Generalized internal-measurement handoff

The common descriptor also separates apparatus-context dependence from a genuine post-body target response. Hold M_τ fixed and suppose one source-owned C^2 post-body functional has regular target block $H_X = D_x^2\mathcal{A}_{\mathrm{post}}$ and mixed block $C_{XO} = D_x D_o \mathcal{A}_{\mathrm{post}}$. Stationarity gives

$$D_o x_* = -H_X^{-1} C_{XO}, \qquad D_o \Pi_a^{\mathrm{occ}} = D_o \Pi_a - D_x \Pi_a H_X^{-1} C_{XO}. \tag{4}$$

Thus a terminal may change directly with context even when $C_{XO} = 0$, and a nonzero target response may be hidden by a terminal null or stable quantizer cell. For hard records the exact comparison is the same-context, clamped-target control. Across a declared terminal family, independently visible target-mediated content is measured by the stacked rank

$$\mathrm{rank} \begin{bmatrix} -D_x \Pi_1 H_X^{-1} C_{XO} \\ \vdots \\ -D_x \Pi_m H_X^{-1} C_{XO} \end{bmatrix}, \tag{5}$$

not by the number of terminal names. This is a conditional response theorem, not a derivation of the physical mixed block, its nonzero occupation, or a component beyond the standard terminal theory; it cannot select or modify the already frozen body.

The paper owns the conditional arrow

$$M_\tau \longrightarrow \text{one post-body source ledger} \longrightarrow \{g, A, \Phi, \rho_O, \text{formal clock response}\}.$$

It does not use terminal residuals to select M_τ and does not construct the body from a readout. The proof order is source ownership, variational derivatives, standard-limit controls, ROOT–TT common preparation and finally source-rank non-identifiability.

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2 Inherited-results ledger

Result	Owner	Status and assumptions	Use here
Body-first descent	Paper I	conditional physical MVP	generative order
Typed incidence	Paper II	finite source theorem	occupied domains
Record recovery	Paper III	exact/approximate conditional theorem	observer terminal
Parent-law realization	Paper IV	present reduct does not entail full packet	claim boundary
Temporal descent	Paper V	conditional clock/causal leaf	time and weak limit
Quantum terminal	Paper VI	conditional complex/effect/QFT packet	Γ_q and ρ_O
Morphology-conditioned readouts	theory hub	common-source assembly for an occupied complete signature	typed terminal projections
Operation-generated occupation	theory hub	cyclic–surjective/root-positive criterion inside a fixed source representation	block nonvanishing
P1 assembly	theory hub	conditional metric/time/radiation ledger	g, null radiation, stress
P3 Abel handoff	theory hub	conditional $Z_F = p_* K_Y$	Maxwell/current sector
Pointed P3 exterior seed	theory hub	conditional orthonormal 256-vector root orbit; full-rank nontracial countermodel	representation boundary
Support–port P3 synthesis	theory hub	commuting support flag, two ordered ports, independent unit and smooth decoder; finite audit	rank-seven source reconstruction
Tau-local composition origin	theory hub; reproved here	exact inside the component-local no-bridge/no-hidden-global-relation class	two-leg record and effect tensor algebra
Component-local P3 joint occupation	owned here	exact inside one once-counted enriched common law; narrow-reduct selection disproved	joint LC1–LC4 and rank-seven status
Scalar-15/type-II completion	theory hub P4 branch	class-relative base–seed reconstruction, common-scale ratio and exact functional RG law; unrestricted branch occupation open	downstream particle-sector handoff

Table 1: Dependency and ownership ledger. No inherited conditional result is promoted to an unrestricted physical parent-law theorem.

For logical self-containment, Appendix A restates every imported premise actually used below. The paper labels in Table 1 are provenance locators, not hidden hypotheses: the conditional arguments can be checked from the restated premises without accepting an unquoted result from a companion manuscript. Several listed papers provide architectural context only and are not logical dependencies of the main theorems.

3 Morphology-conditioned assembly and occupation

The inherited terminals are not treated as independent information channels. Let the stabilized body and observer support determine one finite physical source representation $\mathcal{E}_{*,O}$. No direct-sum

decomposition of this source is assumed. Instead, for the typed role set

$$\mathcal{P} = \{T, G, R, M, Q, A\}, \qquad \mathcal{Y}_{\text{role}} := \bigoplus_{a \in \mathcal{P}} \mathcal{Y}_a, \tag{6}$$

let $R_a : \mathcal{E}_{*,O} \to \mathcal{Y}_a$ be terminal-role maps and $R = (R_a)_a : \mathcal{E}_{*,O} \to \mathcal{Y}_{\text{role}}$ their stacked map. The direct sum is therefore a de-duplicated *role codomain*, not a claim that the source information separates. Common source directions may affect several R_a simultaneously; their overlap is measured by the joint kernel and stacked rank of the R_a.

Let A_O be the body- and observer-conditioned access map and let $Q_O^{(a)}$ be a stable observer-relative resolution map. The operational terminals are

$$Y_O^{(a)} = Q_O^{(a)} R_a A_O \Xi_*, \tag{7}$$

for one occupied source representative Ξ_*. Equality under the specified $Q_O^{(a)}$, rather than a non-transitive distance threshold, defines the resolved readout classes. Thus observer-relative quantization remains part of the full descent and is not replaced by the continuous source geometry.

Proposition 1 (Conditional morphology-conditioned representation)**.** *Assume a stabilized body, one occupied complete source representation $\mathcal{E}_{*,O}$, a source-owned typed ledger map, and stable resolution maps $Q_O^{(a)}$. Then all terminals in Eq.* (7) *are functorial restrictions of one source representative. Their common-source ownership and once-counting follow without terminal-specific gains. Different terminal names do not by themselves imply independent information. This is a representation theorem for an already occupied complete source ledger; it does not establish uniqueness among all admissible source functionals.*

Proof. Body stabilization fixes the domain of A_O. The single map $A_O \Xi_*$ is then mapped by the typed R_a and resolved only afterwards by $Q_O^{(a)}$. Consequently every terminal factors through the same occupied representative and the same access map. Any additional terminal gain would be an extra source datum, contrary to the assumed complete ledger. The conclusion is role-completeness of this construction, not uniqueness under field redefinitions, improvement terms or alternative ultraviolet completions. □

Define the immediate shared descriptor $\Xi_{OS} := A_O \Xi_*$ on the occupied source domain. The complete declared terminal atlas is therefore the typed product

$$\mathfrak{R}_{OS} = \left\langle Q_{OS}^{(a)} \circ R_a \right\rangle_{a \in \mathcal{P}} \circ \Xi_{OS}. \tag{8}$$

This is a product of unlike codomains, not a sum of observables with incompatible units. It includes pure roles and any hybrid map that jointly retains two or more role legs before the final stable quantizer.

The Paper-I common-ancestor theorem makes the assembly falsifiable. A candidate terminal R_{new} belongs to this atlas exactly when it is constant on Ξ_{OS}-fibres; locally on a regular connected-fibre branch,

$$\ker D\Xi_{OS} \subseteq \ker DR_{\text{new}}. \tag{9}$$

The declared stack recovers the whole local ancestor quotient exactly when

$$\ker D\Xi_{OS} = \ker \begin{bmatrix} D(Q^{(T)}R_T) \\ D(Q^{(G)}R_G) \\ D(Q^{(R)}R_R) \\ D(Q^{(M)}R_M) \\ D(Q^{(Q)}R_Q) \\ D(Q^{(A)}R_A) \end{bmatrix}. \tag{10}$$

Strict inclusion means that the terminals share the ancestor but do not reconstruct it; violation of Eq. (9) is an actual bypass. Neither case may be repaired by fitting a terminal gain. Physical no-bypass and completion of the terminal family remain upstream source-occupation obligations.

Let $\mathfrak{A}_{\mathrm{src}}$ be the source-owned operation algebra and Ω_s the occupied source vector, and define $V_s := \mathrm{span}(\mathfrak{A}_{\mathrm{src}}\Omega_s)$.

Definition 1 (Activation and full occupation)**.** *A required terminal role a is activated when $R_a(V_s) \neq \{0\}$. It is fully occupied when*

$$R_a(V_s) = \mathcal{Y}_a. \tag{11}$$

Full occupation implies activation; the converse need not hold.

Assembly does not by itself prove either property. In finite dimension, full occupation of every required role is equivalent to

$$R_a(\mathrm{span}(\mathfrak{A}_{\mathrm{src}}\Omega_s)) = \mathcal{Y}_a \quad \text{for every required } a. \tag{12}$$

Proposition 2 (Operation-generated occupation criterion)**.** *If Ω_s is cyclic for $\mathfrak{A}_{\mathrm{src}}$ on $\mathcal{E}_{*,O}$ and each required R_a is surjective, then Eq.* (12) *holds and every required role is fully occupied. More weakly, let $X_0 \geq 0$ be the occupied root covariance with $\operatorname{supp} X_0 \subseteq V_s$, and let the source-owned path maps L_e preserve the occupied subspace, $L_e(V_s) \subseteq V_s$. Define*

$$X_a := \sum_{e \in \mathcal{P}_a} R_a L_e X_0 L_e^\dagger R_a^\dagger \geq 0. \tag{13}$$

If $R_a L_e X_0^{1/2} \neq 0$ for at least one $e \in \mathcal{P}_a$, then role a is activated, but full rank is not implied.

Proof. Cyclicity gives $V_s = \mathcal{E}_{*,O}$; surjectivity then gives $R_a(V_s) = \mathcal{Y}_a$. In the weaker case, every summand in Eq. (13) is positive semidefinite and

$$\operatorname{Tr} X_a = \sum_{e \in \mathcal{P}_a} \|R_a L_e X_0^{1/2}\|_{\mathrm{HS}}^2. \tag{14}$$

One nonzero path therefore implies $X_a \neq 0$. Because $\mathrm{Ran}(L_e X_0^{1/2}) \subseteq V_s$, its nonzero image lies in $R_a(V_s)$ and hence proves activation. It does not establish Eq. (11). □

Propositions 1 and 2 move the open problem upward. Once the common representation and its source operation law are occupied, separate terminal selection is unnecessary. What remains unproved is that $B_\tau^{\mathrm{phys}} + s_U$ selects this representation, makes Ω_s cyclic with surjective required role maps (or realizes the rooted positive alternative), and chooses the observer access and resolution maps. Algebraic generation, terminal agreement and post-handoff record completeness do not prove that prior physical realization and occupation.

4 The ordered Paper-VII ledger

Let y denote body coordinates, and write $z = (g, A, \Phi)$ for the variational post-body geometry, Abelian gauge and occupied classical matter fields. The quantum preparation is instead a fixed positive trace-one boundary datum $\rho_{\mathrm{in,O}} \geq 0$, $\operatorname{Tr}\rho_{\mathrm{in,O}} = 1$ for the in-in contour. It is not an unconstrained stationary field.

Definition 2 (Once-counted ordered ledger)**.** *The Paper-VII ledger is*

$$\mathfrak{L}_7 = (\mathcal{A}_{\mathrm{pre}}[y; B, s],\ y_*,\ \Gamma_7[g, A, \Phi \mid y_*; \rho_{\mathrm{in,O}}],\ \mathrm{Desc}_{O\leftarrow y_*},\ \mathcal{U})\,, \tag{15}$$

where $D_y\mathcal{A}_{\mathrm{pre}}(y_*) = 0$, the physical body Hessian is positive, and

$$\Gamma_7 = \Gamma_{\mathrm{grav}}[g \mid y_*] + \Gamma_{\mathrm{EM}}[A, g \mid y_*] + \Gamma_{\mathrm{occ}}[\Phi, A, g \mid y_*] + \Gamma_q[A, g \mid y_*; \rho_{\mathrm{in,O}}] + \Gamma_{\mathrm{ct}}[A, g]. \tag{16}$$

Each occupied mode and each counterterm appears in exactly one summand. Observer restriction is evaluated after stationarity and is not varied back against y_*. The state $\rho_{\mathrm{in,O}}$ fixes the CTP preparation; the Euler equations vary g, A and Φ, not $\rho_{\mathrm{in,O}}$. The unit packet $\mathcal{U}$ is shared by every summand.

5 One effective source on each stationary body fiber

The symbol Γ_7 need not be postulated as five unrelated pieces. Let φ collect the occupied quantum fields, and for each declared outcome family let the dimensionless constant probe η_a couple to a normalized POVM-effect insertion satisfying $E_a \geq 0$ and $\sum_a E_a = I$ (hence $0 \leq E_a \leq I$). Let ϑ denote the occupied clock character. The source convention is chosen so that

$$\Gamma_{\mathrm{PC}}[\eta] = \Gamma_{\mathrm{PC}}[0] + \sum_a \eta_a\, \mathsf{E}_a[\varphi], \qquad \langle \mathsf{E}_a \rangle = \operatorname{Tr}(\rho_O E_a). \tag{17}$$

On the already selected stationary body y_* define the doubled closed-time-path influence functional

$$\begin{aligned} \mathcal{I}_{\mathrm{CTP}} := {} & \Gamma_{\mathrm{PC}}[y_*; g^+, A^+, \vartheta^+, \eta^+, \varphi^+] \\ & - \Gamma_{\mathrm{PC}}[y_*; g^-, A^-, \vartheta^-, \eta^-, \varphi^-], \end{aligned} \tag{18}$$

$$\exp\!\left(\frac{i}{\hbar} W_O[g^\pm, A^\pm, \vartheta^\pm, \eta^\pm \mid y_*; \rho_{\mathrm{in,O}}]\right) = \int_{\rho_{\mathrm{in,O}}} \mathcal{D}\varphi^+ \mathcal{D}\varphi^-\, \exp\!\left(\frac{i}{\hbar}\mathcal{I}_{\mathrm{CTP}}\right). \tag{19}$$

The initial boundary condition joins the two branches through the fixed preparation $\rho_{\mathrm{in,O}}$, and the path integral includes the standard final branch identification. The in-in contour is required for real causal expectation values [Schwinger, 1961, Keldysh, 1965]. For every doubled source write $X_c = (X^+ + X^-)/2$ and $X_\Delta = X^+ - X^-$. All derivatives below mean $\delta/\delta X_\Delta$ followed by the physical limit $X_\Delta = 0$, $X_c = X$; differentiating an already identified common branch would instead lose the response by CTP normalization. The construction is a terminal representation of an atemporal ordered functional fibration, not a parent-time history. CTP evolution and observer restriction produce the normalized output state ρ_O used in Eq. (17); neither $\rho_{\mathrm{in,O}}$ nor ρ_O is varied as a local Euler field. Renormalization gives one W_O^{ren} whose physical-limit difference derivatives define

$$p_a = \left.\frac{\delta W_O^{\mathrm{ren}}}{\delta \eta_a}\right|_{\eta=0} = \operatorname{Tr}(\rho_O E_a) \in [0,1], \qquad \langle T^q_{\mu\nu}\rangle_{\mathrm{ren}} = -\frac{2}{\sqrt{-g}}\frac{\delta W_O^{\mathrm{ren}}}{\delta g^{\mu\nu}}, \tag{20}$$

$$\langle J^\mu_q\rangle_{\mathrm{ren}} = -\frac{1}{\sqrt{-g}}\frac{\delta W_O^{\mathrm{ren}}}{\delta A_\mu}, \qquad \mathcal{R}^{\mathrm{clock}}_O = \frac{\delta W_O^{\mathrm{ren}}}{\delta \vartheta}. \tag{21}$$

Consequently $\sum_a p_a = 1$ within each normalized outcome family. Without the POVM and source-coupling conditions, the derivative would be only a generic operator expectation and could not be identified as a Born probability. Here the effect-source renormalization convention is required to preserve the POVM normalization at $\eta = 0$; Eq. (17) is an admitted operational insertion and not a claim that every POVM has a local polynomial field representation. The final expression is schematic until the clock character and its normalization are fixed, but its ownership is unambiguous. The minus sign in the current definition is fixed by the Maxwell convention $\Gamma_{\rm EM} = -(Z_F/4)\int \sqrt{-g}\, F^2$: stationarity then gives $\nabla_\mu(Z_F F^{\mu\nu}) = J^\nu$. We use the same sign convention for the classical current, $J^\mu_{\rm cl} := -(\sqrt{-g})^{-1}\delta\Gamma_{\rm occ}/\delta A_\mu$.

Theorem 1 (Effective common-source construction)**.** *Assume the ordered pre-body solve, a source-owned probe-complete post-body functional $\Gamma_{\rm PC}$, an observer-admissible integration class $\mathcal{C}_O$ with fixed normalized boundary state $\rho_{\rm in,O}$, and a local covariant, anomaly-free renormalization prescription for the symmetries used below. Then Eqs.* (19)–(21)*, together with the classical body and coherent mean-field terms, construct the Paper-VII Γ_7 ledger from one post-body source functional. The construction is complete at the level of the declared physical roles, modulo relocation of a finite local counterterm between the geometric and quantum bookkeeping columns while holding their sum fixed. No uniqueness claim is made across field redefinitions, improvement terms or inequivalent microscopic functionals.*

Proof. Functional differentiation of one $W_O^{\rm ren}$ gives the four displayed terminal owners. Diffeomorphism and gauge invariance give their common Ward covector. The fluctuation Green operator is the inverse of the corresponding gauge-fixed Hessian on the physical quotient (or its prescribed generalized inverse when zero modes remain) and therefore computes response; inserting it again as an action summand would duplicate the fluctuation contribution. A finite local counterterm can be written in either the geometric or renormalized quantum column, but linearity of variation shows that only the sum affects the field equations. Including the same term in both columns doubles its variation. Thus this assumed functional supplies a consistent role ledger, while its harmless counterterm placement convention is not fixed. The argument does not exclude a distinct probe-complete functional with the same terminal equations. □

Scope. Theorem 1 is a construction theorem inside the selected ordered completion. It does not show that the physical base–seed law selects $\Gamma_{\rm PC}$, $\mathcal{C}_O$, its field spectrum or its renormalized continuum measure.

The ordering is structural. If $H_B, H_P > 0$ and the post-body equations have body derivative C, the linear response is lower triangular:

$$\begin{pmatrix}\delta y\\ \delta z\end{pmatrix} = \begin{pmatrix} H_B^{-1} & 0\\ -H_P^{-1}CH_B^{-1} & H_P^{-1}\end{pmatrix}\begin{pmatrix}\delta j_B\\ \delta j_P\end{pmatrix}. \tag{22}$$

Thus morphology conditions the physical terminals, while a later terminal load does not rebuild the body.

Proposition 3 (Simultaneous-action no-go)**.** *For an invertible positive simultaneous Hessian*

$$\mathbb{H} = \begin{pmatrix} H_B & C^\dagger\\ C & H_P\end{pmatrix}, \tag{23}$$

exact vanishing of body response to every pure post-body load implies $C = 0$. Hence nonzero body conditioning and exact body freeze cannot both be encoded by one simultaneous unconstrained Hessian solve.

Proof. Schur inversion gives $\delta y = -H_B^{-1}C^\dagger(H_P - CH_B^{-1}C^\dagger)^{-1}\delta j_P$. If this vanishes for every δj_P, invertibility forces $C = 0$. □

6 Disjoint occupation and stress ownership

Let the occupied mode labels form a disjoint union

$$\mathcal{I}_{\rm occ} = \mathcal{I}_{\rm cl} \sqcup \mathcal{I}_q. \tag{24}$$

Classical coherent modes represented in $\Gamma_{\rm occ}$ are excluded from the state-dependent part of Γ_q; vacuum polarization and local renormalization terms occur only in $\Gamma_q + \Gamma_{\rm ct}$. This disjoint partition is a sufficient transparent ledger, not the only possible field-theoretic organization. A background-field, one-particle- irreducible or two-particle-irreducible construction is also admissible when its mean field, fluctuation determinant and counterterms are derived from one effective action and no full contribution is inserted twice.

Proposition 4 (Duplicate-stress no-go)**.** *If one nonzero occupied mode i is included in both $\mathcal{I}_{\rm cl}$ and $\mathcal{I}_q$ with the same metric variation, its stress contribution is counted twice. No terminal redefinition removes this without changing the metric equation or adding a compensating subtraction.*

Proof. Metric variation is linear. The duplicated summands give $-2(\sqrt{-g})^{-1}\delta(\Gamma_i + \Gamma_i)/\delta g^{\mu\nu} = 2T^{(i)}_{\mu\nu}$. A subtraction merely moves the ownership entry and must therefore be recorded explicitly in $\Gamma_{\rm ct}$. □

7 Joint terminal theorem

The theorem below begins only after the upstream geometry gates have been passed. Gate G4a is existence of a Lorentz-conformal realization of the source-owned cone/order on a regular smooth rank-four leaf; this is open in the present Tau source. Gate G4b is the standard conditional uniqueness of the conformal class under the required distinguishing causality hypotheses. A compatible source-owned density may then fix a representative only under additional normalization assumptions. Local coframes $e^A_{(\alpha)\mu}$ are related on overlaps by $SO^+(1,3)$ transition functions and satisfy

$$g_{\mu\nu}|_{U_\alpha} = e^A_{(\alpha)\mu} e^B_{(\alpha)\nu}\eta_{AB}.$$

Nothing in the conditional standard-leaf theorem asserts a single global coframe, a spin lift, or the DA–ROOT/M4TR same-carrier intertwiner. Those remain separate topology and source-ownership obligations.

Proposition 5 (Conditional standard-leaf form selector)**.** *Assume that the selected readout leaf is a smooth four-dimensional Lorentzian manifold. Suppose the pure geometric action is local, diffeomorphism invariant, metric-only, and has metric Euler equations of at most second derivative order. Suppose also that the Abelian action is local, $U(1)$-gauge invariant, parity even, quadratic in first derivatives of A, contains no derivatives of F, and has a constant positive kinetic coefficient. Then, up to boundary and four-dimensional topological terms, the two actions have the forms*

$$\Gamma_{\rm grav} = \frac{1}{2\kappa}\int d^4x\sqrt{-g}(R - 2\Lambda), \tag{25}$$

$$\Gamma_{\rm EM} = -\frac{Z_F}{4}\int d^4x\sqrt{-g}\, F_{\mu\nu}F^{\mu\nu}, \qquad Z_F > 0. \tag{26}$$

The proposition fixes the action shapes inside this infrared class, but not κ, Λ, Z_F, or physical realization of the class itself.

Proof. The four-dimensional Lovelock classification leaves only the cosmological and Einstein tensors in the local metric-only second-order Euler tensor; the Gauss–Bonnet density is topological in four dimensions [Lovelock, 1971]. Integration gives the Einstein–Hilbert form up to boundary/topological terms and overall normalization. Abelian gauge invariance restricts a local first-derivative action to dependence on $F = dA$. In four dimensions the quadratic Lorentz scalars are $F_{\mu\nu}F^{\mu\nu}$ and $F_{\mu\nu}\widetilde{F}^{\mu\nu}$. Parity removes the second from the dynamical class (and for constant coefficient it is topological), so the first is unique up to Z_F; positivity fixes its sign. These classification arguments neither fix the coefficients nor prove that the parent selects their hypotheses. □

The selector does not rule out higher-curvature gravity, torsion, additional geometric fields, nonlinear electrodynamics or parity-odd sectors; those lie outside its hypotheses. Its content is uniqueness within the declared standard infrared class, not uniqueness among all physical completions.

For the joint theorem adopt Proposition 5, the conditional common-source handoff $Z_F = p_* K_Y$, and a matter functional, quantum measure and renormalization prescription that are diffeomorphism and $U(1)$ gauge invariant and anomaly-free in those symmetries. The handoff removes an independent Maxwell gain inside the declared completion; it does not evaluate p_* or select the occupied representation trace K_Y from the physical parent.

Theorem 2 (Common-descent Einstein–Maxwell–matter assembly)**.** *Under the ordered ledger, disjoint occupation rule, common unit packet and the stated invariances, stationarity of Eq.* (16) *with $\rho_{\mathrm{in,O}}$ held fixed gives*

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa \left(T^{\mathrm{EM}}_{\mu\nu} + T^{\mathrm{cl}}_{\mu\nu} + \langle T^{q}_{\mu\nu} \rangle_{\mathrm{ren}} \right), \tag{27}$$

$$\nabla_\mu (Z_F F^{\mu\nu}) = J^\nu_{\mathrm{cl}} + \langle J^\nu_q \rangle_{\mathrm{ren}}. \tag{28}$$

On the matter and gauge equations, the Ward identities and Bianchi identity imply

$$\nabla^\mu T^{\mathrm{tot}}_{\mu\nu} = 0, \qquad \nabla_\mu J^\mu_{\mathrm{tot}} = 0. \tag{29}$$

No additional gravity, gauge or quantum-backreaction gain is introduced at the terminal stage.

Proof. Equations (27) and (28) are the metric and gauge Euler equations of the single post-body functional. Gauge invariance gives current conservation. Diffeomorphism invariance gives the combined stress Ward identity, including the Lorentz-force exchange between matter and gauge sectors. The divergence of the left side of Eq. (27) vanishes identically, so the equations are mutually consistent. Disjoint occupation and the single-entry counterterm ledger ensure that each contribution is varied once. □

Corollary 1 (Standard weak-field leaf)**.** *For $g_{00} = -(1 + 2\Phi_N/c^2)$, stationary weak fields, nonrelativistic sources and a local patch whose boundary data admit the Newtonian limit, Eq.* (27) *reduces to*

$$\nabla^2 \Phi_N = 4\pi G \rho_{\mathrm{tot}} - \Lambda c^2, \tag{30}$$

where ρ_{tot} is the same once-counted density appearing in the ledger. On scales $L \ll |\Lambda|^{-1/2}$, or for $\Lambda = 0$, the final term is negligible and the ordinary Poisson equation follows. Equation (28) *reduces to the ordinary Maxwell equations after the inherited field normalization. This is standard recovery, not a Tau-specific deviation.*

8 Relation to established frameworks

The use of one renormalized effective action and its metric variation belongs to semiclassical gravity and curved-spacetime QFT, not uniquely to Tau Core [Wald, 1977, Birrell and Davies, 1982, Hollands and Wald, 2001]. The closed-time-path extension and stress-noise cumulants likewise parallel stochastic gravity [Hu and Verdaguer, 2008]. Observer-relative descriptions have independent precedent in quantum-reference-frame constructions [Giacomini et al., 2019], while deriving Einstein dynamics from more primitive thermodynamic data is exemplified by emergent-gravity programs [Jacobson, 1995].

The finite-dimensional algebraic ingredients are also established mathematics rather than Tau-specific discoveries. The exterior realization of Clifford multiplication is standard spin geometry [Lawson and Michelsohn, 1989]; cyclic representations and their unitary uniqueness belong to the GNS/operator-algebra framework [Bratteli and Robinson, 1979]; and special dagger–Frobenius structures provide the finite-dimensional algebraic language for normalized copy/record data [Vicary, 2011, Coecke et al., 2013]. Paper VII's claim is therefore not novelty of these ingredients separately, but their conditional placement in one source-owned, once-counted terminal ledger and the accompanying non-identifiability controls.

Paper VII's narrower Tau-specific contribution is the body-first ordering, the typed morphology-conditioned access and resolution maps, the explicit once-counted cross-terminal ownership ledger, and the finite no-go separating terminal agreement from common provenance. The standard effective-action machinery is used as a consistency leaf; it is not presented as evidence for the Tau parent ontology.

9 Local constraint structure

The covariant equations must reproduce constraints, not only evolution equations. Let n^μ be the unit normal to an observer-compatible spacelike leaf, $h_{\mu\nu} = g_{\mu\nu} + n_\mu n_\nu$ its induced metric, and K_{ij} its extrinsic curvature. With $\rho = n^\mu n^\nu T^{\mathrm{tot}}_{\mu\nu}$ and $j_i = -h_i{}^\mu n^\nu T^{\mathrm{tot}}_{\mu\nu}$, the normal projections of Eq. (27) give, in the stated sign convention,

$$ {}^{(3)}R + K^2 - K_{ij}K^{ij} = 2\kappa\rho + 2\Lambda, \tag{31} $$

$$ D_j(K^{ij} - h^{ij}K) = \kappa j^i. \tag{32} $$

The normal projection of Eq. (28) gives the Abelian Gauss constraint

$$ D_i(Z_F E^i) = \rho_J. \tag{33} $$

The Ward identities propagate these constraints when the classical and quantum matter equations hold. Paper VII therefore imports the standard local constraint architecture from the one joint functional; it does not derive a new constraint algebra or quantize it.

10 Primitive terminal inventory

The assembled packet contains several typed outputs, not several names for the same information:

Family	Primitive terminal content	Derived or composite descendants
Geometry/gravity	g, connection, curvature and constraints	proper time, tidal response, lensing after a null source
Gauge/radiation	A, F, conserved current, two screen polarizations	photon propagation, redshift and optical transport
Matter occupation	occupied localized or propagating solutions Φ	mass, energy, stress and charge records
Quantum operation	ρ, effects, instruments and correctable composition	Born statistics and renormalized quantum stress/current
Observer access	causal orientation, clock and stable restriction	observer-relative terminal classes

Stress-energy, current, redshift and lensing are variational or composite outputs of these families; they are not silently counted as new primitive source directions. Conversely, the symmetric local gravity descriptor cannot stand in for internal gauge charge, quantum phase or occupied matter. A global information-completeness claim must wait for the non-Abelian and fermionic known-terminal completions.

11 Why the standard limit does not prove common provenance

Proposition 6 (Equal-terminal provenance no-go)**.** *The equations* (27)–(29) *do not imply that their coefficients or sectors arise from one Tau source.*

Proof. Construct two source models with the identical numerical effective functional $\Gamma_7[g, A, \Phi]$. In C_{joint} its sector coefficients are images of one common source line and unit map. In C_{sep} take four independent source lines and choose their terminal coefficient maps to have the same numerical images as the four coefficients in C_{joint}. The two models therefore have identical functionals as functions of the terminal fields, so all Euler equations, Ward identities and solution sets coincide. Their source ownership graphs are nevertheless non-isomorphic: one has a common source line and the other has four independent lines. Hence no observation or algebraic manipulation of Eqs. (27)–(29) alone can recover common provenance. This is an identifiability counterexample, not a claim that the independently sourced model is physically preferred. □

This no-go fixes the empirical interpretation: terminal data may test a source-frozen forward ledger, but may not choose its source partition, susceptibilities or unit map.

Proposition 7 (Same-reduct selection no-go)**.** *The currently frozen stationary body, common action line, observer quotient and terminal type signatures do not entail physical occupation of the full effective-source packet of Theorem 1.*

Proof. Hold fixed the body y_*, its Hessian, common units, causal orientation and observer quotient. Add the same typed internal edge carrier $\mathcal{H}_e = \mathbb{C}^2$ to both completions, with effect

$$E_e = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}, \qquad \rho_w = \begin{pmatrix} 1-w & 0 \\ 0 & w \end{pmatrix}, \quad 0 < w < 1, \qquad c_0 = 0, \quad c_1 = 1. \tag{34}$$

ρ_w is positive, has unit trace and is invariant under the diagonal $U(1)$ action. A concrete covariant edge functional is supplied by a charged scalar doublet $\chi \in \mathbb{C}^2$ with

$$\Gamma_e[\chi, g, A] = -\int d^4x\sqrt{-g}\left[g^{\mu\nu}(D_\mu\chi)^\dagger D_\nu\chi + m_e^2\chi^\dagger\chi\right], \qquad D_\mu = \partial_\mu - iqA_\mu I_2. \tag{35}$$

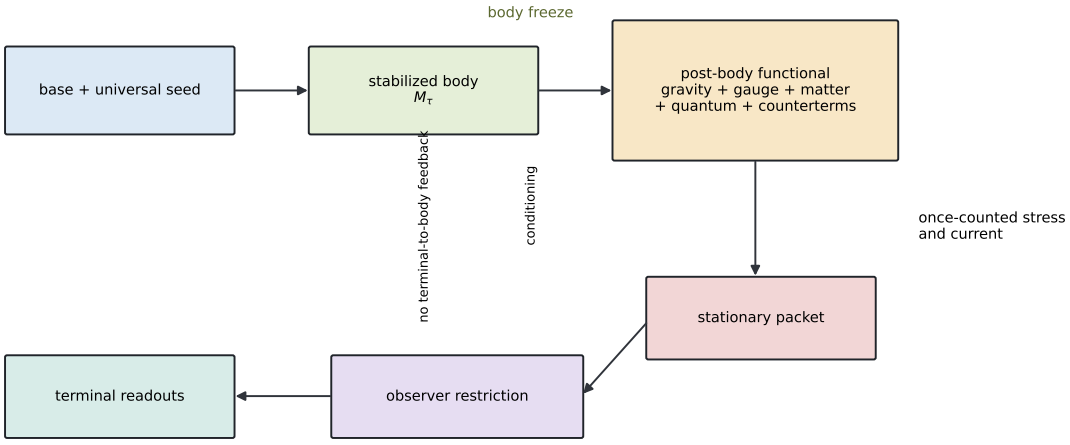


Figure 1: Ordered common-source construction. The body is stabilized before the post-body functional is solved; observer restriction is terminal.

It is diffeomorphism invariant, diagonal-$U(1)$ gauge invariant and has the same positive internal state normalization in both completions. Define

$$\Gamma^{(j)}_{\mathrm{PC}}[\eta_e] = \Gamma_{\mathrm{common}} + \Gamma_e + c_j \eta_e \mathsf{E}_e, \qquad j \in \{0,1\}. \tag{36}$$

At the physical source value $\eta_e = 0$, both completions have exactly the same action, state, body, Hessian, observer quotient, typed carrier, covariance and normalization. The frozen upstream reduct does not include a law selecting the probe coefficient c_j. Nevertheless,

$$\left.\frac{\delta W^{(0)}_O}{\delta \eta_{e,\Delta}}\right|_{\mathrm{phys}} = 0, \qquad \left.\frac{\delta W^{(1)}_O}{\delta \eta_{e,\Delta}}\right|_{\mathrm{phys}} = \mathrm{Tr}(\rho_w E_e) = w > 0. \tag{37}$$

Only C_1 contains the declared nonzero effect-source owner. Since the full effective-source packet includes that owner, the frozen upstream reduct does not entail the complete packet. The result concerns source ownership, not the logically different statement that a realized POVM outcome must have nonzero probability in every state. No stress- or current-difference is claimed because the physical unprobed action and field state are identical. Additional source-owned occupation or an empirically discriminating source-frozen prediction is required. □

This finite counterpair is the stopping result for the origin question at the present source level. It forbids both an unrestricted parent-law promotion of Theorem 1 and another same-level search for a hidden normalization. It does not weaken the conditional forward construction.

Figure 1 summarizes the resulting generative and variational order.

12 Quantum backreaction and the quantum-gravity boundary

The quantum term in Eq. (27) is a renormalized expectation value on a classical body-derived metric, in the standard semiclassical sense [Birrell and Davies, 1982, Wald, 1977]. Local covariance and admissible renormalization follow the usual curved-spacetime QFT framework [Hollands and

Wald, 2001, Brunetti et al., 2003]. The metric can also be driven by a stress-noise second cumulant in a stochastic extension, but this does not quantize geometry.

Paper VII therefore establishes a conditional quantum–gravity *interface*, not quantum gravity. A path integral or operator algebra for the metric, gravitational gauge fixing, a physical inner product and a nonperturbative ultraviolet completion remain absent.

12.1 Observer-dependent terminal mixing and its exact boundary

The common parent descriptor permits a sharper statement than either terminal identity or terminal independence. Let X be a fixed source-owned parent tangent and let

$$A_i := \begin{pmatrix} Q_i \\ G_i \end{pmatrix} : X \longrightarrow Y_{Q,i} \oplus Y_{G,i} \tag{38}$$

be the complete linearized quantum–gravity access of observer O_i, after its stable quantizer, null quotient and calibration have been included.

Proposition 8 (Joint-access mixing)**.** *Two observers resolve the same parent quotient if and only if*

$$\ker A_1 = \ker A_2. \tag{39}$$

In that case there is a unique isomorphism on the realized images, $U_{21} : A_1(X) \to A_2(X)$, such that

$$A_2 = U_{21} A_1. \tag{40}$$

For any three equal-access observers these image transports satisfy $U_{ki} = U_{kj}U_{ji}$, $U_{ii} = I$ and $U_{ij} = U_{ji}^{-1}$.

Proof. If Eq. (40) holds with U_{21} invertible, the two kernels are equal. Conversely, define $U_{21}(A_1 x) := A_2 x$. Equality of the kernels makes this independent of the representative x, and the same construction with the indices exchanged gives its inverse. Uniqueness on $A_1(X)$ is immediate. The groupoid identities follow by uniqueness of the factorizations of A_k through A_i. □

This permits observer-dependent terminal assignment without creating new parent information. On a realized pure quantum sector $E_{Q,1} := A_1(X) \cap (Y_{Q,1} \oplus 0)$, the intrinsic quantum-to-gravity component is $\pi_{G,2} U_{21}|_{E_{Q,1}}$. An arbitrary extension of U_{21} to the ambient direct sum is not unique when $A_1(X)$ is a proper subspace, so its displayed off-diagonal matrix entries are not physical invariants. The rank-two pure-swap witness

$$Q_1 = (1\ 0), \quad G_1 = (0\ 1), \qquad Q_2 = (0\ 1), \quad G_2 = (1\ 0) \tag{41}$$

proves mathematical nonemptiness only.

For unequal full kernels $K_i := \ker A_i$, the maximal common linear quotient and the combined quotient are respectively $X/(K_i + K_j)$ and $X/(K_i \cap K_j)$. Their dimensions are

$$d^{ij}_{\text{shared}} = \operatorname{rank} A_i + \operatorname{rank} A_j - \operatorname{rank} \begin{pmatrix} A_i \\ A_j \end{pmatrix}, \tag{42}$$

$$d^{ij}_{\text{union}} = \operatorname{rank} \begin{pmatrix} A_i \\ A_j \end{pmatrix}. \tag{43}$$

These counts are invariant under invertible observer recalibration and must be evaluated before named terminals are counted as independent.

Information equivalence is not physical sector duality. If a full role exchange is represented by an injective multiplicative intertwiner Φ, then $\Phi([B,C]) = [\Phi(B), \Phi(C)]$. Consequently a genuinely noncommutative quantum operational algebra cannot be fully equivalent in this sense to a commutative classical-gravity algebra. Morita equivalence alone is too weak because it need not preserve element commutators or operational probabilities. More generally, a full and faithful process equivalence that preserves every closed-circuit probability would pull any classical noncontextual simplex model of the gravity fragment back to the quantum fragment. It therefore cannot connect a certified contextual quantum packet to a gravity packet that remains classical under the same independently frozen causal, dimension and memory assumptions.

Raw order dependence is not the required witness: classical transformations, nonlinear backreaction, apparatus memory and unequal preparations can also make $A \circ B$ differ from $B \circ A$. A finite physical promotion requires, in order, (i) common-parent kernel reconstruction, (ii) independent preparation/transformation/measurement equivalences, (iii) rejection of a bounded commutative comparator by contextuality, incompatibility or equivalent tomography, and (iv) one source-frozen map transporting both information and composition. An unrestricted classical hidden-state automaton can encode any finite behavior table, so the comparator bounds are indispensable.

The result generalizes to any typed terminal family $A_i = \bigoplus_\alpha T_i^\alpha$. Equal full kernels give the same image groupoid; pure $\alpha \to \beta$ exchange is tested on the carrier where all competing terminal blocks vanish. Globally, Jacobian kernels must be replaced by equality of the stable quantized fibers. This establishes a conditional architecture in which the joint accessible parent quotient can be observer-invariant while its terminal decomposition is observer-dependent. It does not prove that Nature realizes nonzero cross-terminal mixing, that the metric is quantized, or that quantum and gravity are ontologically identical.

The time block has no privileged information status inside this theorem. If T_O is the clock component of the complete stack A_O, every other terminal factors through it exactly when $\ker T_O = \ker A_O$ locally, or when their stable fibers coincide globally. A scalar rank-one clock therefore cannot determine this paper's higher-rank joint terminal packet. The richer ordered distinguishability carrier may constrain every terminal, but it is upstream of and strictly differently typed from the clock projection.

12.2 Post-body source origin of the access maps

The preceding access maps need not be inserted independently. After body stabilization, let $x \in X$ perturb the frozen parent descriptor and let $z_i \in Z_i$ be the observer–source response. The positive post-body quadratic germ

$$\Phi_i^{(2)}(z_i; x) = \frac{1}{2}\langle z_i, H_i z_i\rangle - \langle z_i, C_i x\rangle, \qquad H_i > 0, \tag{44}$$

has stationary solution $z_i = H_i^{-1} C_i x$. If L_i is the de-duplicated resolved terminal exit, then

$$\boxed{A_i = L_i H_i^{-1} C_i.} \tag{45}$$

Here C_i is the parent/access mixed Hessian of the one post-body source law; Eq. (45) introduces no separate terminal capacity gain and does not place the frozen body derivative back into a simultaneous solve.

The exact access fiber is

$$\ker A_i = C_i^{-1}(H_i \ker L_i)\,. \tag{46}$$

Thus equal observer access requires equality of these preimages. If L_i is injective on the stationary response image, this reduces to $\ker C_i = \ker C_j$; observer stiffness changes sensitivity but not the accessible parent directions. For a lossy exit the full Eq. (46) is required, and changing H_i can rotate a response into or out of the terminal null.

One common positive relation Hessian does not force equal access. Even when

$$H_i = J_i^* K_R J_i, \qquad C_i = J_i^* K_R J_X, \tag{47}$$

different observer embeddings J_i can select different coordinate subspaces and hence different kernels. This gives an exact same-source countermodel to the claim that common provenance itself implies terminal mixing.

A sufficient source-selected route is a covariant observer orbit. If

$$C_j = V_{ji}^{-*} C_i, \qquad H_j = V_{ji}^{-*} H_i V_{ji}^{-1}, \qquad L_j V_{ji} = W_{ji} L_i, \tag{48}$$

with invertible maps on the realized images, substitution into Eq. (45) gives

$$A_j = W_{ji} A_i. \tag{49}$$

The observer transport is then derived from the source packet rather than chosen after terminal inspection. Equation (48) is a sufficient enriched-source condition, not a theorem that physical observers occupy one such orbit. Off-diagonal physical terminal content of W_{ji} remains unselected.

12.3 Representation-overlap selection rule

The covariant orbit law also restricts which off-diagonal blocks can exist. If a physical symmetry group $\mathcal{G}$ acts on realized terminal sectors, source covariance requires every cross-terminal component to lie in

$$W_{ji}^{\beta\leftarrow\alpha} \in \mathrm{Hom}_{\mathcal{G}}(E_i^\alpha, E_j^\beta). \tag{50}$$

Schur's lemma therefore sets the component to zero between inequivalent irreducible sectors. For compact $\mathcal{G}$, after complexification when needed, write

$$E_i^\alpha \simeq \bigoplus_\lambda M_{i,\alpha,\lambda} \otimes V_\lambda. \tag{51}$$

Every covariant cross map is blockwise $w_{ji,\lambda}^{\beta\leftarrow\alpha} \otimes I_{V_\lambda}$ and obeys

$$\operatorname{rank} W_{ji}^{\beta\leftarrow\alpha} \le \sum_\lambda \min\{\dim M_{i,\alpha,\lambda}, \dim M_{j,\beta,\lambda}\} \dim V_\lambda. \tag{52}$$

Representation overlap is necessary, not sufficient: source normalization, orientation, gauge descent and operational-law transport remain required.

This rule makes the ROOT–TT packet below the first current eligible quantum–gravity pair. Its two-mode coherence and TT stress carry the same oriented spin-two screen type. A scalar probability or spin-one current cannot mix covariantly into that pure spin-two terminal without an additional symmetry-breaking carrier. The ROOT–TT identity still proves only common-preparation transport into semiclassical stress, not invertible quantum–gravity role duality.

12.4 Coupling is weaker than an intrinsically hybrid terminal

For a positive joint quadratic packet

$$K_{QG} = \begin{pmatrix} A & C \\ C^* & B \end{pmatrix}, \qquad A > 0, \quad B > 0, \tag{53}$$

$C \neq 0$ proves coupling in the declared quantum/gravity splitting. It does not prove absolute hybridity: an unrestricted triangular congruence produces $A \oplus (B - C^* A^{-1} C)$. A meaningful hybrid claim must therefore freeze the admissible transformations from terminal type, gauge, units, quantizer and operational composition before inspecting C.

Under invertible block-preserving transformations, $C \mapsto S_Q^* C S_G$ and $\operatorname{rank} C$ is invariant. Under independent metric-preserving terminal frames, the singular values of

$$M = A^{-1/2} C B^{-1/2} \tag{54}$$

are invariant and satisfy $0 \leq \sigma_k(M) < 1$ by positivity. Their nonzero ordered set is a dimensionless typed quadratic-hybrid spectrum. It measures normalized coupling, not physical quantum–gravity unification.

Physical hybridity additionally requires one occupied once-counted mixed source block, survival through gauge/null/quantizer descent, resolved process nonfactorization against a frozen typed comparator and preservation of both operational signatures. The current ROOT–TT map passes representation eligibility and common preparation but does not yet supply this occupied joint Hessian or the operational nonfactorization theorem. It is therefore a coupled candidate, not an established hybrid terminal.

The exact deterministic identity is in fact more restrictive. Write its retained linear form as $t = Kq$ and let $Sigma_Q > 0$ be the ROOT covariance or static susceptibility on its support. Then

$$\Gamma_{QT} = \begin{pmatrix} \Sigma_Q & \Sigma_Q K^* \\ K\Sigma_Q & K\Sigma_Q K^* \end{pmatrix} = \begin{pmatrix} I \\ K \end{pmatrix} \Sigma_Q \begin{pmatrix} I & K^* \end{pmatrix}. \tag{55}$$

Consequently $\operatorname{rank} \Gamma_{QT} = \operatorname{rank} \Sigma_Q$ and the TT-given-ROOT Schur residual vanishes. Every nonzero canonical correlation is unity on the supported image. Thus the presently proved ROOT–TT relation is a redundant graph of one shared preparation, lying on the semidefinite boundary of the strictly positive family in Eq. (53); it is not itself a two-sector hybrid proof.

The minimal covariance extension $t = Kq + \eta$ with $\langle \eta q^* \rangle = 0$ and $N := \langle \eta \eta^* \rangle > 0$ has Schur residual exactly N and is strictly positive. This does not close the physical argument: N is a new underived TT source/noise and the same packet has a classical Gaussian realization. A nonarbitrary continuation must instead reconstruct the connected mixed susceptibility of one renormalized source functional,

$$C_{QT} = \left. \frac{\delta^2 W[h_{TT}, \eta_Q]}{\delta h_{TT}\, \delta \eta_Q} \right|_0, \tag{56}$$

with contact and subtraction terms frozen, and must then prove quotient survival and operational nonfactorization. Equation (59) alone supplies none of these additional conclusions.

12.5 Two-mode common-preparation theorem

The preceding interface can be made more specific without quantizing the metric. Fix an oriented observer Sachs screen and an occupied two-mode QFT restriction with annihilation operators a_m, a_{m+2}. Let

$$C_{ij} := \langle a_i^\dagger a_j \rangle, \qquad N_O := \operatorname{Tr} C > 0, \qquad \rho_{\mathrm{ROOT}} := \frac{C}{N_O}. \tag{57}$$

The two screen weights differ by two, so their coherence transforms in the same spin-two representation as the complex screen stress $T_+ + iT_\times$. Denote the once-counted normal-ordered STF overlap of the two occupied modes by $K_{m,m+2}$. It is not a free terminal gain. Let u_i be the source-normalized mode functions, let $e_+^\mu = (e_1^\mu + ie_2^\mu)/\sqrt{2}$ be the oriented complex screen vector, and let f_O be a normalized observer smearing window. If $\mathcal{T}_{\mu\nu}[u_i^*, u_j]$ denotes the polarized bilinear appearing in the once-renormalized Hilbert stress derived from the same QFT action, define

$$K_{ij}[u, f_O] := \int d\mu_O \, f_O \, e_+^\mu e_+^\nu \mathcal{T}_{\mu\nu}[u_i^*, u_j]. \tag{58}$$

Thus the action, mode normalization, screen, smearing and subtraction scheme determine K_{ij}. The common-preparation identity is

$$\boxed{\langle : T_+ + iT_\times : \rangle = K_{m,m+2} C_{m,m+2} = N_O K_{m,m+2} (\rho_{\mathrm{ROOT}})_{m,m+2}.} \tag{59}$$

Proposition 9 (Conditional ROOT–TT common preparation)**.** *Assume (i) the physical restriction occupies a common, oriented screen pair whose weights differ by two; (ii) the relevant state is restricted to a number-preserving two-mode second-moment sector, so anomalous moments $\langle a_i a_j \rangle$ and $\langle a_i^\dagger a_j^\dagger \rangle$ vanish or are independently projected out; (iii) all other mode, vacuum and polarization contributions to the selected STF component are orthogonal or included once in the declared subtraction; (iv) the ROOT one-particle reduced state and the stress expectation are formed from the same source-owned QFT preparation; (v) normal ordering or renormalization is entered once in the source ledger; and (vi) $K_{m,m+2} \neq 0$, with $K_{m,m+2}$ calculated by Eq.* (58) *from the declared source preparation, screen and subtraction prescription. Then Eq.* (59) *follows and, after $K_{m,m+2}$ is source-frozen, admits no additional independent terminal-specific gain within this restricted sector.*

Proof. The ROOT one-particle reduced state is the normalized one-particle coherence matrix by Eq. (57). Under assumptions (ii)–(iii), projection of the normal-ordered quadratic stress onto the selected spin-two component leaves $K_{m,m+2}\langle a_m^\dagger a_{m+2} \rangle$. Substitution of $C = N_O \rho_{\mathrm{ROOT}}$ gives Eq. (59). Introducing an additional gain after $K_{m,m+2}$ is fixed would assign two calibrations to one occupied preparation and would violate the once-counted source ledger. The formula does not select the mode pair, screen or subtraction prescription and does not prove that the resulting overlap is nonzero. Changing those source-side inputs changes the calculated $K_{m,m+2}$; it does not authorize a post-hoc terminal gain. □

The use of the unnormalized matrix C, rather than only ρ_{ROOT}, also closes the quantum-induced part of the P1 gluing. Let $J_C : \delta s \mapsto \delta C$, let π_{12} extract the real and imaginary parts of $C_{m,m+2}$, let M_K denote real multiplication by the source-frozen complex number $K_{m,m+2}$, and let L_{P1}^{TT} be the retarded linear response from the screen-TT stress plane to the P1 geometry rows. Then

$$\boxed{J_{P1}^{(Q)} = L_{P1}^{\mathrm{TT}} M_K \pi_{12} J_C,} \qquad \ker J_C \subseteq \ker J_{P1}^{(Q)}. \tag{60}$$

If $K_{m,m+2} \neq 0$ and L_{P1}^{TT} is injective on the occupied TT plane, this restriction has real rank two. This is a factorization theorem for the quantum-induced stress row, not for the complete P1 terminal. The distinction is essential: $J_Q(x, y, z) = (x, y)$ and $J_{P1}(x, y, z) = (x, y, z)$ have a rank-two common submap although $(0, 0, 1) \in \ker J_Q \setminus \ker J_{P1}$. Thus local ROOT–TT rank two does not prove the full-carrier zero-cokernel condition. Conversely, replacing C by normalized ρ_{ROOT} alone loses this factorization under occupation-scale variations: $\delta C = \epsilon C$ gives $\delta \rho_{\mathrm{ROOT}} = 0$ but a nonzero TT-stress variation whenever the coherence and overlap are nonzero.

On the physical pure-state ROOT tangent the existing body-first staging makes this statement stronger. With observer and radiative boundary data frozen, the direct Green row, stabilized body stress and direct ROOT-to-occupied-matter term have zero variation. Therefore

$$\boxed{J_{P1}|_{\mathcal{V}_Q} = J_{P1}^{(Q)}|_{\mathcal{V}_Q} = L_{P1}^{\mathrm{TT}} M_K \pi_{12} J_C|_{\mathcal{V}_Q}.} \tag{61}$$

Thus Eq. (60) is the full local P1 factorization on the two-real-dimensional post-body ROOT tangent, while the counterexample above continues to exclude a global conclusion on the complete source domain.

The local result can nevertheless be removed exactly from the remaining global rank calculation. Choose the source-metric orthogonal splitting $\mathcal{E}_{\mathrm{phys}} = \mathcal{E}_R \oplus \mathcal{V}_Q$ and write $J_Q = (C_R, D)$ and $J_{P1} = (A_R, \iota D)$, where D is injective on the two-dimensional ROOT tangent and $\iota = L_{P1}^{\mathrm{TT}} M_K \pi_{12}$. If P_D projects onto $\operatorname{im} D$, define

$$\mathcal{R}_{\mathrm{rest}} x = \big((I - P_D) C_R x, (A_R - \iota C_R) x\big), \qquad x \in \mathcal{E}_R. \tag{62}$$

Then projection to $\mathcal{E}_R$ gives

$$\boxed{\ker \begin{pmatrix} J_Q \\ J_{P1} \end{pmatrix} \simeq \ker \mathcal{R}_{\mathrm{rest}}, \qquad \operatorname{rank} \begin{pmatrix} J_Q \\ J_{P1} \end{pmatrix} = 2 + \operatorname{rank} \mathcal{R}_{\mathrm{rest}}.} \tag{63}$$

Indeed, $C_R x + D y = 0$ is soluble exactly when $(I - P_D) C_R x = 0$ and then has the unique solution $y = -D^{-1} C_R x$; the P1 equation becomes $(A_R - \iota C_R) x = 0$. Thus the already closed ROOT tangent contributes exactly two ranks. The unresolved differential content is confined to the rest-domain transverse quantum component and any independent P1 component. This is an exact staged reduction, not physical realization of the complete base–seed packet or a replacement for finite-support and algebraic faithfulness tests.

Nor do the current marginal data determine $\operatorname{rank} \mathcal{R}_{\mathrm{rest}}$. Take $\mathcal{E}_R = \mathbb{R}^4$, $D = \iota = I_2$, $C_R = (I_2\ 0)$ and respectively

$$A_0 = (I_2\ 0), \qquad A_1 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \end{pmatrix}, \qquad A_2 = (0\ I_2). \tag{64}$$

All three controls have the same faithful local ROOT block, raw ranks and marginal Grams $J_Q J_Q^\dagger = J_{P1,k} J_{P1,k}^\dagger = 2 I_2$, but $\operatorname{rank} \mathcal{R}_{\mathrm{rest},k} = k$. Thus the remaining rank requires the rest-domain mixed P1–quantum incidence. It is calculated by the common-source mixed-Gram formula once the full packet is physically selected; separate marginal closure does not select it.

The inherited common rank-four body theorem removes a larger local block. Let the local source split be $X_B \oplus H_{\mathrm{hid}}$ and write the retained rank-four observer descriptor and P1 differential as

$$J_4 = (A\ B), \qquad J_{P1} = (P\ Q), \qquad A : X_B \xrightarrow{\sim} Y_4. \tag{65}$$

Full local qubit and Lorentz retention makes A invertible. Defining

$$\Delta_{\mathrm{hid}} = Q - P A^{-1} B \tag{66}$$

and applying exact block elimination gives

$$\boxed{\ker \begin{pmatrix} J_4 \\ J_{P1} \end{pmatrix} \simeq \ker \Delta_{\mathrm{hid}}, \qquad \operatorname{rank} \begin{pmatrix} J_4 \\ J_{P1} \end{pmatrix} = 4 + \operatorname{rank} \Delta_{\mathrm{hid}}.} \tag{67}$$

Hence the P1 map factors uniquely through the observer/ROOT descriptor on X_B; no selector is missing on that common core. If the descent is body-block preserving, then $B = 0$ and the remaining defect is simply Q. This result does not determine the hidden-fibre block, identify the local rank-four packet with the complete 64-real-dimensional P3 descriptor, or prove physical occupation of either completion.

The complete P3 comparison can be reduced on the same source split. Write $J_{P3} = (R\ S)$ and remove its already counted common-core component:

$$\Gamma_{\mathrm{hid}} := S - RA^{-1}B. \tag{68}$$

The same invertible block operations give

$$\operatorname{rank}\begin{pmatrix} J_4 \\ J_{P3} \\ J_{P1} \end{pmatrix} = 4 + \operatorname{rank}\begin{pmatrix} \Gamma_{\mathrm{hid}} \\ \Delta_{\mathrm{hid}} \end{pmatrix}. \tag{69}$$

Consequently the hidden P1 defect is represented by P3 exactly when

$$\boxed{\ker \Gamma_{\mathrm{hid}} \subseteq \ker \Delta_{\mathrm{hid}}} \quad \Longleftrightarrow \quad \Delta_{\mathrm{hid}} = L_{P1\leftarrow P3}\Gamma_{\mathrm{hid}} \quad \text{on } \operatorname{im}\Gamma_{\mathrm{hid}}. \tag{70}$$

This is a source-fibre condition, not a target-rank condition. Indeed, $\Gamma_{\mathrm{hid}}(x, y) = x$ is onto its one-dimensional P3 target while $\Delta_{\mathrm{hid}}(x, y) = y$ detects a P3-null source direction. Thus complete P3 operator-state separation does not by itself close the P1–P3 gluing. Source-fibre injectivity of Γ_{hid} would be sufficient, while the weaker kernel inclusion in Eq. (70) is necessary and sufficient. Neither condition is presently selected by the physical base–seed law.

The remaining branch choice is represented by one canonical source-fibre obstruction,

$$\mathcal{O}_{P1|P3} := \Delta_{\mathrm{hid}}\big|_{\ker \Gamma_{\mathrm{hid}}}. \tag{71}$$

Rank–nullity and Eq. (69) give the exact novelty identity

$$\boxed{\operatorname{rank}\begin{pmatrix} J_4 \\ J_{P3} \\ J_{P1} \end{pmatrix} - \operatorname{rank}\begin{pmatrix} J_4 \\ J_{P3} \end{pmatrix} = \operatorname{rank}\mathcal{O}_{P1|P3}.} \tag{72}$$

Thus $\mathcal{O}_{P1|P3} = 0$ is equivalent to complete P1 factorization through P3 after the common core has been counted. If the obstruction is nonzero, its source direction is P1-visible, so it is neither a joint readout null nor an unloaded Schur-neutral presentation auxiliary; source-minimal quotienting cannot remove it. The current parent laws admit both branches and do not select the value of this obstruction.

The common source Hessian also gives a direct evaluation formula. Let $K_{\mathrm{hid}} > 0$ be its Schur complement after the common rank-four block is eliminated and define

$$\widehat{\Gamma} = \Gamma_{\mathrm{hid}}K_{\mathrm{hid}}^{-1/2}, \qquad \widehat{\Delta} = \Delta_{\mathrm{hid}}K_{\mathrm{hid}}^{-1/2}, \qquad \Pi_{\ker\widehat{\Gamma}} = I - \widehat{\Gamma}^{+}\widehat{\Gamma}. \tag{73}$$

Then

$$\boxed{\operatorname{rank}\mathcal{O}_{P1|P3} = \operatorname{rank}\left(\widehat{\Delta}\Pi_{\ker\widehat{\Gamma}}\right).} \tag{74}$$

Equivalently, form the marginal and mixed Grams of $\widehat{\Gamma}$ and $\widehat{\Delta}$. If $\mathcal{C}_{\Gamma\Delta}$ is their support-normalized mixed Gram, the principal-angle theorem gives

$$\operatorname{rank}\mathcal{O}_{P1|P3} = \operatorname{rank}\Delta_{\mathrm{hid}} - \#\{j : \sigma_j(\mathcal{C}_{\Gamma\Delta}) = 1\}. \tag{75}$$

Thus one physically selected common incidence packet computes the obstruction without an additional terminal coefficient. The Hessian alone does not fix it: distinct typed incidences over the same positive metric can realize zero or nonzero obstruction.

This obstruction also separates two notions of completeness that must not be conflated. Let $N_{\mathrm{phys}} \subset H_{\mathrm{hid}}$ be the source null annihilated by every retained terminal, so in particular $N_{\mathrm{phys}} \subseteq \ker \Delta_{\mathrm{hid}}$. If P3 is complete on physical source fibres modulo this joint null, then

$$\ker \Gamma_{\mathrm{hid}} \subseteq N_{\mathrm{phys}} \subseteq \ker \Delta_{\mathrm{hid}}, \tag{76}$$

and therefore

$$\boxed{\text{P3 physical-source completeness} \Longrightarrow \mathcal{O}_{P1|P3} = 0 \Longrightarrow J_{P1} = L_{P1\leftarrow P3} J_{P3}} \tag{77}$$

after the common core and joint null have been removed. Conversely, a nonzero obstruction provides a P3-null source direction that P1 still sees; it therefore certifies failure of P3 source-fibre completeness even when P3 is onto and informationally complete on its quantum operator-state target. Equation (77) is a conditional completeness criterion, not a proof that the physical base–seed parent selects the source-complete branch.

The existing seed-chain/Hodge construction fixes a genuine subpacket but does not close this selection problem. It supplies a nonzero nilpotent incidence on the occupied chain image. On a source split $E_{\mathrm{phys}} = E_{\mathrm{chain}} \oplus H_{\mathrm{ext}}$, however, fix the same positive metric and take

$$\Gamma_{\mathrm{hid}} = (1,0,0), \qquad \Delta_0 = (2,0,0), \qquad \Delta_1 = (2,1,0). \tag{78}$$

The two completions have the same nonzero chain restriction and the same onto P3 map, while $\Delta_0|_{\ker \Gamma_{\mathrm{hid}}} = 0$ and $\operatorname{rank}(\Delta_1|_{\ker \Gamma_{\mathrm{hid}}}) = 1$. Thus the proved chain incidence must be retained and may not be fitted again, but it does not select the complete terminal-typed extension J_*. That remaining extension is the physical representation of the primitive relation presentation on H_{ext}, not another Hodge coefficient or terminal gain.

One selected chain block would extend uniquely if a source-owned typed-isometric transport groupoid made its orbit span E_{P3} and respected all stabilizer relations. This sufficient route is unavailable in the current typed ledger: the two-role observer–instrument defect and the three-role photon–matter defect cannot belong to one type-preserving source orbit. Nor does cyclicity of the represented P3–record GNS carrier replace this condition. For example, the defining $M_2(\mathbb{C})$ representation is cyclic from e_1 because $M_2(\mathbb{C})e_1 = \mathbb{C}^2$, whereas a typed diagonal source group preserves the line $\mathbb{C}e_1$. Thus target-algebra cyclicity does not imply typed source-orbit generation. At least one source-owned inter-orbit relation or separately occupied orbit representative is still required for the full extension.

This remaining relation has an exact finite closure criterion. Decompose the typed source carrier into its source orbits, $V_r = \bigoplus_\alpha V_\alpha$, let $\mathcal{A}_0$ be the star-algebra generated by the fixed intra-orbit representations, and let B contain the proposed source-owned inter-orbit bridge blocks. Define

$$\mathcal{A}_{\mathrm{br}} = \mathrm{alg}^*(\mathcal{A}_0, B). \tag{79}$$

For fixed orbit metrics, two relative gluings preserve all generated source relations exactly when their relative unitary lies in $U(\mathcal{A}'_{\mathrm{br}})$. Assuming $G_{\mathrm{decl}} \subseteq U(\mathcal{A}'_{\mathrm{br}})$, the gluing is unique modulo the declared gauge group precisely when

$$\boxed{U(\mathcal{A}'_{\mathrm{br}}) = G_{\mathrm{decl}}.} \tag{80}$$

This is an internal sufficiency-and-uniqueness theorem, not an existence theorem for B. In the two-scalar-orbit control, the two orbit projectors alone have a two-dimensional commutant and leave an independent relative phase. Adding one nonzero off-diagonal bridge and its dagger generates $M_2(\mathbb{C})$ and reduces the commutant to the common scalar line. This bridge is required only when the declared gauge is the common scalar. Under the inclusion assumed above, the actual obstruction is the excess commutant

$$d_{\text{excess}} = \dim U(\mathcal{A}'_{\text{br}}) - \dim G_{\text{decl}}. \tag{81}$$

If the two diagonal phases are instead the declared two-sector gauge-superselection center, the unbridged model already has $d_{\text{excess}} = 0$; an off-diagonal bridge would then erase a declared superselection direction and is inadmissible without an upstream refinement of the gauge quotient.

This qualification does not close the P3 source problem. The existing superselection theorem concerns the quark/lepton multiplet split $\{Q, u^c, d^c\}$ versus $\{L, e^c\}$, whereas the present no-go separates relation defects with typed supports $\{Q, O\}$ and $\{R, M, G\}$. These are not the same decomposition. Moreover, Q denotes the ROOT/quantum role in the P3 packet but the quark-doublet module in the multiplet theorem; equality of the glyph is not a typed intertwiner. No current theorem transfers the former gauge center to the latter defect carrier. The current base–seed packet therefore fixes neither the relevant G_{decl} nor, if a positive excess remains, the required bridge. The adopted P3 handoff is post-body data and cannot be used backwards to generate this upstream source relation.

More precisely, a target gauge action U_g on a carrier H induces a gauge action on the relation-defect carrier V_r only after a source-owned injective intertwiner $\iota_r : V_r \to H$ is supplied and $U_g \operatorname{im} \iota_r \subseteq \operatorname{im} \iota_r$. Then and only then the pullback is unique:

$$\iota_r \gamma_r(g) = U_g \iota_r, \qquad G_{\text{decl}} = \gamma_r(G_{\text{phys}}) / \ker \gamma_r. \tag{82}$$

The odd-Clifford central projectors act on $\Lambda^\bullet E_{P3}$, the two-point record projectors on R_2, and the quark/lepton center on multiplet modules. None acts on V_r through a currently derived ι_r. Their common two-sector count and equal trace weights therefore do not determine G_{decl} on the typed defect orbits.

There is nevertheless a source-internal positive route. The primitive faithful GNS completion already contains a typed injection $\iota_r : V_r \to \mathcal{A}_{\text{src}}/N_{\omega_s}$. Let $G^{\text{typed}}_{\text{GNS}}$ be its state-preserving star-automorphisms that preserve the image and role typing. Their GNS implementation is unitary, so the gauge-null subgroup relevant to the gluing theorem is determined by

$$\boxed{G^{\text{GNS}}_{\text{decl}} = \gamma_r(G^{\text{typed}}_{\text{GNS}}) \cap U(\mathcal{A}'_{\text{br}}).} \tag{83}$$

Consequently no independent gauge quotient is needed after that complete GNS source packet is selected. What remains open is precisely its physical base–seed selection and whether its typed stabilizer exhausts the physical gauge group.

The canonical exterior–record construction does not prove that selection. For its already supplied positive role metric K_*, the creation–contraction operator $C_{K_*}(u) = \epsilon(u) + \iota_{K_*(u,-)}$ obeys the Clifford anticommutator. Cyclicity of the normalized exterior trace therefore gives

$$\boxed{\operatorname{Re} \tau_{\text{ext}} \left(C_{K_*}(u)^* C_{K_*}(v) \right) = K_*(u, v).} \tag{84}$$

This proves exact compatibility and the absence of a free GNS metric gain, but it does not derive K_*: the contraction leg already contains that metric. A reverse use of Eq. (84) would be circular. The noncircular source target is instead a premetric typed presentation

$$g_r : V_r \longrightarrow \mathcal{A}_{P3}, \qquad j_0 : \mathcal{A}_{P3} \hookrightarrow \mathcal{C}_{16}, \tag{85}$$

fixed without K_*. If j_0 is unital and injective and $j_0 g_r$ is injective on the physical defect quotient, then

$$G_0(u,v) = \operatorname{Re} \tau_{16}\left[(j_0 g_r(u))^* (j_0 g_r(v))\right] \tag{86}$$

is positive definite. The once-counted relation action can then derive its Hessian, GNS carrier, trace weights and conditional gauge pullback from this one packet. The present manuscript proves the implication, not physical base–seed occupation of Eq. (85).

The frozen P3 Clifford candidate gives one noncircular realization of this target. Take the typed basis (B, O, R, U, Q, M, G), let g_r map it to the seven self-adjoint generators of $\mathrm{Cl}_7(\mathbb{C})$, and let j_0 be their faithful normalized Jordan–Wigner embedding as the first seven generators of $\mathrm{Cl}_{16}(\mathbb{C})$. Their defining anticommutator and the normalized trace give

$$\operatorname{Re} \tau_{16}\left[(j_0 g_r(u))^* (j_0 g_r(v))\right] = \langle u, v \rangle_{\mathbb{R}^7}. \tag{87}$$

Hence this declared completion jointly supplies the faithful GNS packet, identity defect metric, equal odd-Clifford central weights and the conditional gauge pullback without importing a Hessian. It proves nonemptiness, not unrestricted parent-law realization. Unequal rescalings $z_a \mapsto \lambda_a z_a$ retain the envelope, dagger and role labels while changing the trace metric to $\mathrm{diag}(\lambda_a^2)$. The current base–seed laws contain no typed full-role operator-square incidence that selects the normalized Clifford relation over this counterfamily.

There is, however, a strictly weaker conditional selector than a transitive symmetry among the physically different roles. Suppose all seven typed P3 lines have a source-owned degree-one lift into one primitive exterior calculus. Let ϵ_a and ι_a be its creation and contraction maps, with

$$\{\epsilon_a, \epsilon_b\} = 0, \qquad \{\iota_a, \iota_b\} = 0, \qquad \{\iota_a, \epsilon_b\} = \delta_{ab} I, \qquad \epsilon_a^\dagger = \iota_a. \tag{88}$$

If the self-adjoint response is role-local and has no independent degree-one bypass, then $C_a = \lambda_a(\epsilon_a + \iota_a)$. If, in addition, every occupied primitive slot is counted once in the same source action unit,

$$\tau(C_a^2) = 1 \qquad \text{for every } a, \tag{89}$$

then $\lambda_a^2 = 1$ and Eq. (88) gives

$$\boxed{\{C_a, C_b\} = 2\delta_{ab} I.} \tag{90}$$

The remaining signs are primitive-frame reflections and do not alter the metric or Clifford algebra. Thus common primitive cost removes the unequal role-local rescalings without identifying the terminal roles. Role-locality is essential: two unit-cost generators related by an oblique mixture can have a nonzero cross anticommutator.

This reduces but does not close unrestricted parent-law realization and occupation. The existing PUSC3 law owns exterior closure and common unit cost on its declared finite local primitive source algebra. The current base–seed derivation does not yet construct a complete typed lift of (B, O, R, U, Q, M, G) into that one primitive slot class or exclude a separate degree-one bypass on its image. The open physical statement is therefore this typed primitive lift, rather than an arbitrary metric coefficient or a new target carrier.

Nor can the closest existing off-diagonal constructions be substituted by matrix shape alone. The rooted R_{root}/TC–CVP1 Hessian acts between pair-difference and pair-common W_2 occurrence modules, not between the P3 defect orbits. The wall–observer/internal separator candidate leaves distinct multiplet blocks disconnected, while the existing EACER, seed-normal and BCCT squared-Hodge carriers have no typed full-role incidence from E_{P3}. These are domain failures, not missing numerical gains.

The later P3 reconstruction reduces the premise without changing this no-go. On the declared enriched completion, the canonical exterior–record realization fixes the P3 matrix carrier, its joint record occupation, the minimal cyclic/GNS representation and the central trace weights. Contractible unloaded Schur-neutral presentation directions are quotient-equivalent. These facts remove four apparent selectors, but they begin after the typed positive P3 quotient and its metric have been supplied. They do not derive the base–seed tangent map J_*, its P1 and observer restrictions, the exact closure of a loaded or typed-visible complementary internal sector, or an isolated rank-two ROOT cluster of the full observer-shorted Hessian. Thus the remaining source packet is finite and explicit:

$$\boxed{\begin{array}{c} J_*, \quad \text{exact complement closure}, \quad \operatorname{gap}_{\mathrm{ROOT}}(K_{*,O}) > 0, \\ \text{physical realization of the enriched completion} \end{array}} \tag{91}$$

Once this packet is source-owned, Eq. (118) evaluates the mixed incidence and Eq. (63) evaluates the stacked rank. A new P3 matrix realization, gain or terminal-specific selector cannot supply the missing value.

In particular, the later quantum factor-designation result does not close this source problem. Its observer-event incidence is defined only on an already adopted quantum event quotient: under exact recovery it selects one logical M_2 embedding, after which canonical descent fixes the complementary M_4 realization algebra. That map is neither J_* nor the complete body–observer incidence into the joint P1–P3 recovery ideal. The domains and codomains differ, so factor designation cannot imply the missing rest-domain rank or full typed surjectivity.

The mode-pair premise and the nonzero-overlap premise admit a sharper finite certificate. Let P_O be the occupied rank-two projector on the selected one-particle sector and let J_{scr} be the self-adjoint generator of rotations of the oriented observer screen. In a J_{scr} eigenbasis write

$$P_O C P_O = \begin{pmatrix} p_m & c \\ c^* & p_{m+2} \end{pmatrix}. \tag{92}$$

Proposition 10 (Finite ROOT–TT activation certificate)**.** *Within the declared number-preserving quadratic matter sector, the selected rank-two cluster is a covariantly closed source of a nonzero two-component TT mean of real rank two if and only if*

$$[P_O, J_{\mathrm{scr}}] = 0, \qquad \operatorname{spec}(P_O J_{\mathrm{scr}} P_O) = \{m, m+2\}, \qquad c \neq 0, \qquad K_{m,m+2} \neq 0. \tag{93}$$

For the opposite helicity the spectral gap is -2.

Proof. Global phase invariance leaves the quadratic coherences $C_{mn} = \langle a_m^\dagger a_n \rangle$. Under a screen rotation they carry character $n - m$, whereas $T_+ + iT_\times$ carries character $+2$. Orthogonality of $SO(2)$ characters therefore removes every off-diagonal coherence except $n - m = 2$ (or -2 for the conjugate helicity). Projector commutation makes these weights well defined inside the occupied cluster. Equation (59) then reduces the TT mean to the complex product $K_{m,m+2}c$. Multiplication by a nonzero complex number has real matrix $\left(\begin{smallmatrix} \Re K & -\Im K \\ \Im K & \Re K \end{smallmatrix}\right)$, whose determinant is $|K|^2$; hence the response has real rank two exactly when both factors are nonzero. If projector invariance fails, the selected subspace is not a covariantly closed two-mode source. Within an invariant cluster, failure of the weight, coherence or overlap condition makes the selected quadratic TT mean vanish. □

Four controls delimit the statement. First, a single geometric-optics null ray has stress $Ik \otimes k$ and zero STF projection on its own Sachs screen; non-collinearity relative to a common screen is

therefore essential. For a ray with transverse components (p_1, p_2),

$$T_+ + iT_\times = \frac{I}{2}(p_1 + ip_2)^2, \tag{94}$$

whose real Jacobian has rank two away from $p_1 = p_2 = 0$. Second, ρ_{ROOT} alone cannot determine the stress magnitude: preparations with the same normalized state and different N_O have different stress. Third, an incoherent classical anisotropic stress can be physically relevant, but it does not establish the coherence-controlled identity above. Fourth, outside the declared number-preserving sector the full covariance packet, including anomalous moments, replaces C; Eq. (59) must not be extrapolated to a generic QFT state.

There is also an explicit nonzero witness for the final condition in Eq. (93). On a patch where an isolated ray has fixed nonzero transverse direction $p_1 + ip_2$ relative to the common screen, positive intensity I and a nonnegative smearing function f_O with nonzero integral, its smeared STF coefficient is proportional to

$$K_{\mathrm{ray}} = \frac{1}{2}(p_1 + ip_2)^2 \int d\mu_O \, f_O I \neq 0. \tag{95}$$

This proves that the admissible nonzero-overlap class is nonempty. It does not show that an arbitrary multimode packet avoids destructive cancellation, or that the physical parent selects this witness.

Proposition 9 is a conditional common-source bridge, not an unrestricted parent-realization theorem. This paper does not prove that the physical base–seed system occupies the required non-collinear $\Delta m = 2$ pair, that every ROOT carrier has this two-mode realization, or that geometry is quantized.

12.6 Minimal invariant activation potential

The four activation conditions need not be independent switches. Let p be a weight-one complex transverse coordinate on the common screen and c the weight-two relative coherence. Under screen rotations $p \mapsto e^{i\theta}p$ and $c \mapsto e^{2i\theta}c$, the lowest-total-degree analytic real mixed invariant is unique up to one coefficient and complex conjugation, $\mathrm{Re}(c^* p^2)$. Higher-order invariants are not excluded. Consider

$$\mathcal{V}(c, p) = r_c|c|^2 + u_c|c|^4 + r_p|p|^2 + u_p|p|^4 - 2\lambda \,\mathrm{Re}(c^* p^2), \tag{96}$$

where $u_c, u_p, r_c > 0$, $r_p < 0$ and $\lambda \in \mathbb{R} \setminus \{0\}$.

Proposition 11 (Conditional minimal activation)**.** *Every global minimizer of Eq.* (96) *has* $p_* \neq 0$ *and* $c_* \neq 0$*, with the relative phase locked by* $\arg c_* = 2 \arg p_*$ *up to the sign of* λ*. If the once-counted STF overlap is* $K(p) = \kappa p^2$ *with* $\kappa \neq 0$*, the full ROOT–TT activation certificate holds.*

Proof. The positive quartics make $\mathcal{V}$ coercive. At $p = 0$, its unique minimum is $c = 0$ with value zero. Along $c = 0$, however, $|p|^2 = -r_p/(2u_p)$ gives $\mathcal{V} = -r_p^2/(4u_p) < 0$, so a global minimizer cannot have $p = 0$. At $p \neq 0$, the c^* variation evaluated at $c = 0$ is $-\lambda p^2 \neq 0$, so $c = 0$ is not stationary. Phase minimization gives the doubled-angle lock. The square p^2 has screen weight two and therefore selects the $\Delta m = 2$ coherence, while $p_* \neq 0$ and $\kappa \neq 0$ give nonzero STF overlap. □

This proposition closes activation only inside the declared minimal completion. The current physical reduct does not determine $r_p < 0$, $\lambda \neq 0$ or $\kappa \neq 0$. In particular, $\lambda = 0$ is an exact same-sector control with $c_* = 0$.

12.7 Source-loaded activation from an occupied Abelian field

The spontaneous branch above is not the only realization. If a nonzero transverse coordinate p_{src} is already supplied by the occupied source ledger, minimize only

$$\mathcal{V}_{p_{\mathrm{src}}}(c) = r_c|c|^2 + u_c|c|^4 - 2\lambda\,\mathrm{Re}(c^* p_{\mathrm{src}}^2), \qquad r_c, u_c > 0. \tag{97}$$

Proposition 12 (Source-loaded coherence selector)**.** *If $p_{\mathrm{src}} \neq 0$ and $\lambda \neq 0$, Eq. (97) has a unique minimizing amplitude and it obeys*

$$(r_c + 2u_c|c_*|^2)c_* = \lambda p_{\mathrm{src}}^2, \qquad c_* \neq 0. \tag{98}$$

Its phase has screen weight two. Therefore this branch closes the coherence condition of Proposition 10 without assuming $r_p < 0$.

Proof. For fixed phase, the radial factor on the left side of Eq. (98) is strictly increasing from zero to infinity. It consequently has exactly one positive solution for every nonzero right side, and phase minimization aligns c_* with $\lambda p_{\mathrm{src}}^2$. Since p_{src} has screen weight one, its square has weight two. □

The mixer in Eq. (97) is also fixed once geometry and stress are varied in the same functional. With the complex STF convention $q_h = h_+ + ih_\times$ and $q_T = T_+ + iT_\times$, write

$$q_h = \beta p_{\mathrm{src}}^2, \qquad q_T = Kc. \tag{99}$$

Proposition 13 (Metric-variation ownership of the mixer)**.** *Suppose the matter stress is defined by variation of the same once-counted matter functional used for the ROOT preparation,*

$$\delta_g \Gamma_{\mathrm{matter}} = -\frac{1}{2}\int d^4x\sqrt{-g}\,T^{\mu\nu}h_{\mu\nu}. \tag{100}$$

On the common screen, Eq. (99) generates a mixed term

$$-g_{\mathrm{TT}}\,\mathrm{Re}(\beta K^* c^* p_{\mathrm{src}}^2), \qquad g_{\mathrm{TT}} > 0, \tag{101}$$

where g_{TT} is fixed by the declared screen and smearing normalization. After the common source phase is fixed,

$$\lambda = \frac{g_{\mathrm{TT}}}{2}|\beta K|. \tag{102}$$

Thus the mixer is nonzero exactly when the geometric TT pullback and the once-counted stress overlap are both nonzero; it is not an independent cross-terminal gain.

Proof. For two real STF screen tensors, their contraction is proportional to $\mathrm{Re}(q_T^* q_h)$. Substitution of Eq. (99) into Eq. (100) gives Eq. (101); the positive proportionality factor contains only the frozen screen, measure and normalization conventions. A common phase choice puts its coefficient in the real form used in Eq. (97), yielding Eq. (102). Adding another multiplier would count the same metric variation twice. □

The Abelian field supplies a canonical candidate for p_{src}. Define

$$I_F = \frac{1}{2}F_{ab}F^{ab}, \qquad J_F = \frac{1}{2}F_{ab}({}^*F)^{ab}, \qquad \Delta_F = I_F^2 + J_F^2. \tag{103}$$

For $\Delta_F > 0$, the non-null Faraday field has an unordered pair of distinct principal null directions $\{[k_+], [k_-]\}$ [Wylleman et al., 2020]. Given observer u, line-of-sight $q = u + n$ and its oriented screen projector $S_{(u,q)}$, set

$$p_\pm = \frac{S_{(u,q)}k_\pm}{-u \cdot k_\pm}, \qquad \chi_{\mathrm{dir}} = |\varepsilon_S(p_+, p_-)|. \tag{104}$$

These expressions are invariant under rescaling or exchange of the null representatives.

Proposition 14 (Conditional Faraday directional selector)**.** *On the non-null Abelian branch,* $\chi_{\mathrm{dir}} > 0$ *supplies two linearly independent common-screen directions and hence a nonzero transverse source for the loaded coherence selector. For a non-principal observer, writing* $k_\pm/(-u\cdot k_\pm) = u + s_\pm$ *gives*

$$\chi_{\mathrm{dir}} = |n \cdot (s_+ \times s_-)|, \tag{105}$$

so the directional condition fails only for sight lines on the great circle spanned by s_+ *and* s_-*. A principal observer, a null field* $\Delta_F = 0$*, and an aligned sight line are exact zero controls.*

Proof. The non-null electromagnetic classification gives two distinct principal null lines. Screen projection and normalization produce Eq. (104); its two-dimensional determinant is nonzero exactly for linearly independent projections. Direct projection into $n^\perp$ gives the scalar triple product Eq. (105). For a non-principal observer $s_+ \times s_- \neq 0$, and its orthogonal sight lines form one great circle. For a principal observer $s_- = -s_+$, while a null field has one repeated principal null line, so both controls give zero. □

This removes an arbitrary directional-pair choice *after* occupation of the non-null Abelian branch. It does not prove that the Tau parent selects that branch, that the QFT preparation lift identifies its support with the two-mode ROOT cluster, or that every observer lies outside the exact control sets.

12.8 Faithful common-completion occupation

The hub's common-completion criterion can be stated directly in the present ledger. Let $K_* > 0$ be the true-null-quotiented body Hessian and let $R_{\mathrm{rec}}^{\mathrm{dedup}}$ be the de-duplicated primitive recovery incidence into the required finite recovery ideal. Define

$$G_*^{\mathrm{rec}} = R_{\mathrm{rec}}^{\mathrm{dedup}} K_*^{-1} (R_{\mathrm{rec}}^{\mathrm{dedup}})^\dagger, \qquad \rho_*^{\mathrm{rec}} = \frac{G_*^{\mathrm{rec}}}{\operatorname{Tr} G_*^{\mathrm{rec}}}. \tag{106}$$

Proposition 15 (Conditional faithful occupation reduction)**.** *If* $R_{\mathrm{rec}}^{\mathrm{dedup}}$ *is surjective, then* ρ_*^{rec} *is faithful on the required recovery ideal and its cyclic GNS representation is faithful and unique up to unitary equivalence. In particular, for every nonzero positive central projector* P_A *of the required Abelian ideal,*

$$w_A := \operatorname{Tr}(\rho_*^{\mathrm{rec}} P_A) > 0. \tag{107}$$

Thus Abelian block occupation requires no independent terminal occupation bit. If a stable ROOT projector P_O *has nonzero weight in the same recovered QFT preparation* Γ_O*, then*

$$\rho_{\mathrm{ROOT}} = \frac{P_O \Gamma_O P_O}{\operatorname{Tr}(P_O \Gamma_O P_O)} \tag{108}$$

and the ROOT state and Hilbert stress are automatically restrictions of one preparation.

Proof. Surjectivity makes G_*^{rec} positive definite on the recovery ideal, so its normalized state is faithful. A faithful state is strictly positive on every nonzero positive element, giving Eq. (107). Faithfulness of the state makes the corresponding finite-dimensional GNS representation faithful, while the standard GNS uniqueness theorem fixes its cyclic realization up to a unitary intertwiner. Finally, Eq. (108) is the normalized compression of the same Γ_O whose metric variation defines the stress; hence no second ROOT preparation is introduced. □

This closes sector occupation and common-state ownership *conditional* on one surjective recovered branch. It does not imply $\Delta_F > 0$ at every spacetime point: faithfulness of the Abelian ideal is a support statement, whereas null versus non-null Faraday type is a local state/solution property. The remaining unrestricted physical problem is therefore finite: the enriched parent law must realize a unique stable branch with surjective $R_{\mathrm{rec}}^{\mathrm{dedup}}$, and the selected observer context must meet the non-null and nondegenerate-screen conditions above. The current reduct does not entail that surjectivity; Proposition 7 supplies its deficient control.

12.9 Exact source-rank reduction

The hub also supplies an exact factorization of the recovery incidence. Let $X_{\mathrm{src}}^{\mathrm{lin}}$ contain the body-owned and irreducible observer–source inputs after the true-null quotient, let $E_{\mathrm{req}}^{\mathrm{lin}}$ be the declared linearized recovery carrier, and write

$$J_{\mathrm{phys}}^{\mathrm{lin}} : X_{\mathrm{src}}^{\mathrm{lin}} \longrightarrow E_{\mathrm{MVP}}^{\mathrm{lin}}, \qquad R_{\mathrm{rec}}^{\mathrm{lin}} = U_{\mathrm{rec}} D_{\Sigma} J_{\mathrm{phys}}^{\mathrm{lin}}. \tag{109}$$

Here D_{Σ} is the ordered lower-triangular response map and U_{rec} implements source-owned deduplication.

Proposition 16 (Exact source-rank criterion)**.** *Suppose D_{Σ} and U_{rec} are isomorphisms on their declared physical quotients and $H_{\mathrm{src}} > 0$ is the once-counted source Hessian. Then*

$$\operatorname{rank} R_{\mathrm{rec}}^{\mathrm{lin}} = \operatorname{rank} J_{\mathrm{phys}}^{\mathrm{lin}}, \qquad \ker R_{\mathrm{rec}}^{\mathrm{lin}} = \ker J_{\mathrm{phys}}^{\mathrm{lin}}, \quad H_{\mathrm{phys}} = (J_{\mathrm{phys}}^{\mathrm{lin}})^{\dagger} H_{\mathrm{src}} J_{\mathrm{phys}}^{\mathrm{lin}}. \tag{110}$$

Consequently the realized branch is strictly stable and its differential recovery has zero cokernel if and only if $J_{\mathrm{phys}}^{\mathrm{lin}}$ is an isomorphism. Nonzero projection into every typed role is not sufficient.

Proof. Left composition by the two isomorphisms preserves rank and kernel, proving the first two identities. Moreover $\langle x, H_{\mathrm{phys}} x\rangle = \langle J_{\mathrm{phys}}^{\mathrm{lin}} x, H_{\mathrm{src}} J_{\mathrm{phys}}^{\mathrm{lin}} x\rangle$, so positivity of H_{src} makes strict stability equivalent to injectivity of $J_{\mathrm{phys}}^{\mathrm{lin}}$. Surjectivity of the linearized recovery map is equivalent to surjectivity of $J_{\mathrm{phys}}^{\mathrm{lin}}$. Both properties hold exactly when the latter map is an isomorphism. □

A full-column-rank map from a lower-dimensional source quotient may have a nonzero component in every required role and yield $H_{\mathrm{phys}} > 0$, while still having nonzero cokernel. This is the exact stable-but-incomplete control. Conversely, the hub's occupied-P3 result supplies a nonempty faithful ROOT-edge subrepresentation inside the declared enriched MVP, but a faithful subrepresentation is not global surjectivity of $J_{\mathrm{phys}}^{\mathrm{lin}}$. Thus the open physical statement is no longer a list of terminal choices: it is whether the enriched base–seed/body plus irreducible observer–source law selects one complete differential incidence with zero kernel and zero cokernel, while the same packet covers the finite record support and acts faithfully on the physical algebra quotient.

Write

$$J_{\mathrm{phys}}^{\mathrm{lin}} = [J_{BS}, J_{OS}^{\mathrm{irr}}], \qquad P_{BS} = J_{BS} J_{BS}^{+}, \qquad C_{OS} = (I - P_{BS}) J_{OS}^{\mathrm{irr}}. \tag{111}$$

Proposition 17 (Observer–source complement rank)**.** *On a source-frozen metric chart,*

$$\operatorname{rank} J_{\mathrm{phys}}^{\mathrm{lin}} = \operatorname{rank} J_{BS} + \operatorname{rank} C_{OS}. \tag{112}$$

If $N_{\mathrm{lin}} = \dim E_{\mathrm{MVP}}^{\mathrm{lin}}$, differential recovery has zero cokernel exactly when

$$\operatorname{rank} C_{OS} = N_{\mathrm{lin}} - \operatorname{rank} J_{BS}. \tag{113}$$

Proof. Split the target orthogonally as $\operatorname{im} J_{BS}$ and its orthogonal complement. Column operations remove the component of J_{OS}^{irr} inside the first summand; its remaining component is C_{OS} and has orthogonal image. The ranks therefore add, and Eq. (113) is the zero-cokernel condition. □

The hub's WR-T113/WR-T117 construction evaluates this complement on the retained six-row PBSE subsystem. Its base plus four access rows have rank five, while the irreducible observer–source row is outside their span and raises the rank to six. Thus, whenever this subsystem embeds faithfully into the Paper-VII linearized carrier,

$$\operatorname{rank} C_{OS} \geq 1. \tag{114}$$

This rules out a purely base–seed factorization on that retained subsystem. It does not determine N_{lin} or prove Eq. (113) for the complete carrier. Moreover differential rank cannot replace finite record coverage or algebraic faithfulness; the three conditions are jointly required for full typed recovery.

12.10 De-duplicated primitive recovery dimension

The broad P1 incidence R_{P1} already includes the common body–matter and causal/time lanes, together with metric/gravity, radiation and observer-screen differentials. On the source-minimal nonzero P1 branch its rank is four. Let R_Q be the real linearization of the occupied two-mode ROOT incidence on the same frozen source quotient and define the source-space row projector

$$P_{P1} = R_{P1}^{+} R_{P1}. \tag{115}$$

Proposition 18 (De-duplicated P1–ROOT rank)**.** *For the adopted P1–ROOT–gauge packet,*

$$R_{\mathrm{rec}}^{\mathrm{lin}} = \begin{pmatrix} R_{P1} \\ R_Q(I - P_{P1}) \end{pmatrix}, \qquad \operatorname{rank} R_{\mathrm{rec}}^{\mathrm{lin}} = 4 + r_{Q\perp}, \tag{116}$$

where

$$r_{Q\perp} = \operatorname{rank}_{\mathbb{R}} R_Q(I - P_{P1}) = \operatorname{rank}_{\mathbb{R}} R_Q - \dim_{\mathbb{R}}(\operatorname{row} R_Q \cap \operatorname{row} R_{P1}), \qquad 0 \leq r_{Q\perp} \leq 4. \tag{117}$$

An invertible gauge transformation on the retained charged/polarization carrier changes the presentation but adds no recovery rank.

Proof. The second block in Eq. (116) lies in the orthogonal complement of the row space of R_{P1}, so the two ranks add. Restricting the orthogonal projection $I - P_{P1}$ to $\operatorname{row} R_Q$ gives Eq. (117) by rank–nullity. The real dimension of the two-mode complex ROOT packet is at most four. Left composition by an invertible gauge map preserves both row-space dimension and stacked rank. □

Separate marginal P1 and ROOT metrics do not determine $r_{Q\perp}$: equal identity marginal Grams admit row-space intersections of real dimension four, two or zero, and hence increments $0, 2, 4$. The missing datum is not another gain but the common-source mixed Gram. With a positive source Hessian K_*, observer access A_O, isolated occupied ROOT projector P_Q and faithful ROOT restriction C_Q, it is

$$G_{P1,Q} = R_{P1} K_*^{-1} A_O^{\dagger} P_Q C_Q^{\dagger}. \tag{118}$$

After normalizing by the two marginal Grams, the singular values are the cosines of the principal angles between the whitened row spaces. Therefore

$$r_{Q\perp} = \operatorname{rank} R_Q - \#\{j : \sigma_j(G_{P1}^{-1/2} G_{P1,Q} G_Q^{-1/2}) = 1\}. \tag{119}$$

Thus the differential recovery rank is confined to $4 \le \operatorname{rank} R_{\mathrm{rec}}^{\mathrm{lin}} \le 8$, but no unique value follows until the enriched physical source selects the common packet $(K_*, R_{P1}, A_O, P_Q, C_Q)$. The PBSE transverse row proves genuine observer–source novelty on its retained subsystem; it must not be added again to Eq. (116) unless an independent embedding outside the broad P1 row space is proved.

Proposition 19 (Conditional seed-presentation selector)**.** *Let the universal seed own a complete finite involutive presentation $\{r_\alpha\}$ of the de-duplicated primitive role algebra and let*

$$\mathcal{A}_{\mathrm{rep}}(j) = \frac{\mathcal{A}_*}{2} \sum_\alpha \|r_\alpha(j)\|_{G_\alpha}^2. \tag{120}$$

Assume: (i) its zero set is one typed-unitary orbit represented by Π_{phys}; (ii) the relation derivative is injective on the gauge/true-null quotient, so $K_* = D_r^* G D_r > 0$; (iii) R_{P1} and A_O are fixed typed projections of the same represented tangent $J_* = D\mathcal{E}_{\Pi_{\mathrm{phys}}}$; and (iv) the shorted operator

$$K_{*,O} = (A_O K_*^{-1} A_O^*)^{-1} \tag{121}$$

has an isolated stable occupied complex rank-two cluster with Riesz projector P_Q and faithful represented ROOT restriction C_Q. Then, up to the same typed-unitary gauge, the seed representation selects

$$(K_*, R_{P1}, A_O, P_Q, C_Q), \qquad R_Q = C_Q P_Q A_O, \tag{122}$$

and Eqs. (118)–(119) *follow without an additional terminal coefficient.*

Proof. The single zero orbit fixes Π_{phys} up to typed unitary equivalence. Differentiation of its represented generator evaluation and the fixed role projections give R_{P1} and A_O, while the Hessian of Eq. (120) at the zero gives K_*. Functional calculus of the isolated cluster fixes P_Q covariantly, and faithfulness fixes C_Q up to a terminal frame. Such a frame cancels from the ranks and mixed-Gram principal angles, proving Eq. (122) and the claim. □

This reduces five apparent inputs to two physical premises, but does not prove them. The current seed grammar does not yet supply a complete uniquely represented P3 (or full eight-role) involutive presentation, and the physical observer-shorted Hessian has not been proved to contain the required occupied isolated cluster. An occupied complete ROOT edge alone is insufficient: its internal restriction can be held fixed while the complementary spectrum is chosen either separated or interleaved. The physical spectral proposition is therefore undetermined, not false.

13 Technical-paper handoffs

Technical Paper VII-B proves, inside its declared finite completion class, the rank-seven support–port reconstruction, Tau-local subsystem composition, pointed exterior carrier, operator-square selector and class-minimal enriched P3 realization. Technical Paper VII-C separately owns the scalar-15/type-II source preimage, occupation no-go and conditional activation, common-scale ratio and exact functional-RG evaluation boundary. Neither imported result is a theorem that the unrestricted physical base–seed parent occupies its completion.

The separator-side ledger is inherited in its compressed Paper-II form. A single coherent source datum $\mathfrak{J}_* = (\Pi_*, \varphi_*)$ determines the mixed relation ideal, multiplicity pairing, relative $P3$–$P4$ gluing and faithful retained-block occupation up to the declared gauge. The present terminal assembly introduces none of these as an independent coefficient or fit parameter. It also cannot infer $\mathfrak{J}_*$

from terminal agreement: the current narrow source non-entailment and the incompleteness of the global terminal family remain upstream physical boundaries.

The upstream source quotient is likewise understood in its complete form:

$$\mathfrak{c}_{B,s} = (q_B, \beta_B).$$

It retains both the decoded universal seed and every independently physical base modulus. Only $\ker D\mathfrak{c}_{B,s}$ is removed as realization redundancy. Consequently terminal coefficients inherited from β_B are neither adjustable gains nor gauge artifacts, and this assembly paper does not infer their values from downstream terminal fits.

The temporal/quantum interface has an analogous typed boundary. CCOC owns a rank-three relative map $A_M : V_{OS}^{\mathrm{rel}} \to S_B$, and OGAS places its contact row and occurrence row on one signed rank-four section. This is enough for the local orientation imported by the clock terminal, but not for a full source-owned connector

$$\Xi : T(\mathbb{CP}^1_s \times \mathbb{CP}^1_b)|_U \longrightarrow E_T \oplus E_S \tag{123}$$

or for equality of the projective and body volume measures. Conditional on a split-preserving $\Xi_{\gamma_T} = \gamma_T \oplus A_M$, fixed oriented product measures and $|\det A_M| = g_{FFL}^3$, exact measure matching uniquely yields

$$\gamma_T = g_{FFL}^{-3}. \tag{124}$$

Thus the common terminal ledger may not introduce a separate clock/body gain, but neither can its standard-limit agreement be used backward to infer physical ownership of Ξ.

Corollary 2 (Conditional three-level series closure)**.** *Assume the common-source occupation and once-counted terminal premises proved in this paper and, independently, the occupied class-minimal P3 realization of Technical Paper VII-B. Then the declared series completion is closed at three separate levels: common source, algebra/representation and terminal response. No blockwise gain is introduced by composing the two packets.*

Proof. The present paper supplies the common source and terminal levels. Technical Paper VII-B supplies the algebra/representation level using the same declared unit and typed source ledger. Composition therefore adds no new normalization. The result remains conditional on both papers' physical-realization premises. □

14 Decision ledger

Statement	Status	Boundary
One once-counted functional generates effect, metric, gauge and clock-source responses	proved conditionally	body and source packet are frozen first
Einstein–Maxwell–matter equations and Ward/Bianchi identities	recovered inside the standard leaf	action shapes and anomaly freedom are premises
Classical and quantum stress are not double counted	proved by ownership ledger	depends on declared renormalization bookkeeping
ROOT and TT arise from one two-mode preparation	proved in the declared sector	ROOT is a one-particle reduced state, not full tomography
ROOT closes all remaining P1 source directions	not established	exact mixed-Gram/source-rank obstruction remains
Standard-limit agreement proves common Tau provenance	disproved	explicit same-reduct counter-models
Legacy terminal law survives the ontology refinement	proved conditionally	$L = U \circ \Xi_{OS}^{\mathrm{cont}}$; equal rank alone is insufficient
Unrestricted parent realizes and occupies the full joint packet	open	not inferable from finite terminal agreement
Signed rank-four section identifies projective and body volume measures	not established	a full source-owned Ξ is still required; if supplied, $\gamma_T = g_{FFL}^{-3}$ is fixed rather than fitted
EOCC actuality supplies an additional 4D dark-sector stress	not derived	requires a separately sourced and conserved terminal stress map; proto-order is not itself energy
Quantized geometry or full Standard Model	not derived	outside Paper VII-A

15 Occupied-body input inherited from parent realization

The joint terminal ledger is evaluated only after occupied body support is frozen. Upstream finite-local analysis shows that every positive seed-unloaded split sector has zero response, whereas every stable loaded or cross-coupled direction belongs to generated support. Consequently no inert finite-local extension supplies an additional terminal gain or stress term here. This does not derive the gravity–matter–gauge packet from the unrestricted base–seed law and does not establish ambient exhaustivity.

Conditionally on the enriched one-source premise, Paper IV's operational identity makes that occupied support the seed-generated physical quotient. Terminal assembly cannot enlarge it: an inert ambient summand contributes no stress, charge or clock term, while any non-null terminal difference at fixed complete base–seed key signals a second source or an incomplete upstream signature. The joint standard leaf does not prove the premise.

The upstream words "base", "seed" and "body" use Paper I's minimal formation basis. A physical zero-source member anchors the base response; the seed is a response-distinct base-relative source class; and the body is a stable selected nonneutral class. Without the zero-source anchor the base–seed split has an exact common-refactorization freedom. Gravity, matter and gauge terminal agreement therefore cannot identify that split or prove physical occupation. Paper VII-A begins only after the occupied body has been frozen.

The assembly does not assume that the physical base is homogeneous or that the seed owns all initial inhomogeneity. Those assignments define one conditional upstream branch. Inhomogeneous-base, mixed base–seed and spontaneous-breaking origins remain unresolved and cannot be selected

by gravity–matter–gauge terminal agreement.

In the homogeneous equivariant unique-response specialization, the inherited split is sharper. If Q projects onto occurrence-inhomogeneous modes, $Qb_0 = 0$ and $[K_B, Q] = 0$, then subtraction of the base-only solution gives

$$\boxed{Qm_*(s) = K_B^{-1} Q J_U(s).} \tag{125}$$

The complete transverse body input to this paper is then seed-induced, while the base supplies its response scale and stabilization. Paper VII-A does not derive those upstream premises. An inhomogeneous base load, a stiffness that mixes homogeneous and transverse sectors, or a nonunique spontaneously broken body law can reproduce a nonconstant body without the same seed attribution. No agreement among the gravity, matter and gauge terminals can remove that source non-identifiability retrospectively.

The inherited body statement must also be separated from the observed 4D contrast. Linearize a terminal exit $\mathcal{R}_{4,O}$ about the base-only solution and set $\Delta m_* = K_B^{-1} Q J_U(s)$. Then

$$Q_{4,O} \Delta Y_{4,O} = Q_{4,O} D_m \mathcal{R}_{4,O}\, K_B^{-1} Q J_U(s) + Q_{4,O} D_a \mathcal{R}_{4,O}\, \Delta a_O. \tag{126}$$

The first term is an inherited body mode jointly dependent on seed forcing and base susceptibility; the second is a possible observer–source/readout contrast. A unique deterministic equivariant exit cannot generate inhomogeneity from a homogeneous body, whereas an observer-relative correlated quantized exit can do so conditionally. Paper VII-A derives neither that latter physical kernel nor a decomposition of observed terminal structure into base, seed and readout shares.

Global-law proof obligation

The joint terminal assembly inherits occupied body support. Under the source-fidelity/Hodge parent law, every positive-Schur additional noncontractible sector has zero effective source and hence zero occupation. This prevents an inert richer topology from supplying a hidden terminal gain. It does not establish that Nature owns that global law: a componentwise zero-jet preserves the entire local assembly while changing the global minimum. Nature-level unification therefore requires independent source or empirical evidence for global-law ownership.

Charged Hodge state and the common-cone boundary

The inherited Hodge law has a sharper consequence for this terminal assembly. On a charged carrier $\mathcal{A}_{\mathrm{ch}} = M_4(\mathbb{C}) \otimes \mathcal{A}_{\mathcal{M}}$, assume that the complete source Hodge operator is geometrically Clifford-natural:

$$[K_s, X \otimes I] = 0 \qquad \forall X \in M_4(\mathbb{C}).$$

Then $K_s = I_4 \otimes K_{\mathcal{M}}$, and its standard full-support spectral state is

$$\rho_{\mathrm{ch}} = \frac{I_4}{4} \otimes \rho_{\mathcal{M}}.$$

Thus the charged geometric evaluation is root-free and the photon and observer terminals use one inherited cone; Paper VII-A may not add a separate spin-root selector or cone-matching gain. This is a class-relative input from the occupied Hodge completion. A faithful exponential state by itself does not suffice, and neither terminal agreement nor the present unrestricted reduct proves that the physical universe occupies this class.

Joint effect–stress source selection and phase gluing

The once-counted functional fixes a coordinate-invariant cross-terminal quantity once its typed sources are frozen. On the true-null-quotiented descriptor tangent, let $K > 0$ and let

$$\ell_Q = D_z D_\eta W[a], \qquad \ell_G = D_z D_h W[f].$$

The normalized overlap

$$\sigma_{QG} = \frac{|\ell_Q K^{-1} \ell_G^*|}{\sqrt{(\ell_Q K^{-1} \ell_Q^*)(\ell_G K^{-1} \ell_G^*)}}$$

contains no terminal gain when (W, K, z_*, a, f) is fixed upstream. The unsourced action and body do not fix it: equal-reduct completions can vary the two source incidences and realize the full interval $[0, 1]$.

The conditional gluing itself needs no additional quantum–gravity primitive. If the occupied Hessian commutes with the parent complex structure and the whitened observer map intertwines it,

$$[K, J] = 0, \qquad RJ = J_O R,$$

then $C_O = RK^{-1}R^*$, its supported inverse and every isolated spectral effect projector commute with J_O. Phase invariance of the same functional makes its coframe stress derivative and effect derivatives parts of one variational packet. This is precisely the zero phase-pairing-defect condition inherited from the EBOA common-bundle theorem. A source-owned $\mathbb{Z}_4$-closed occupied orbit is sufficient, but the current positive metric and unsigned path data do not select that occupation. Thus the paper adds a conditional gluing theorem and a physical occupation boundary, not a quantum-gravity detection or universal mixing constant.

16 Source-frozen full-4D standard excess and fixed joint score

The joint terminal ledger admits one precise scoring consequence. Let the linearization of one co-registered solder–transport source packet on the observer-visible and unresolved parent legs be J_H, J_V, D_H, D_V. With the common relation and morphology weights,

$$\begin{aligned} K_{HH} &= J_H^\dagger W_\rho J_H + D_H^\dagger W_M D_H, \\ C &= J_H^\dagger W_\rho J_V + D_H^\dagger W_M D_V, \\ K_{VV} &= J_V^\dagger W_\rho J_V + D_V^\dagger W_M D_V. \end{aligned}$$

For $K_{VV} > 0$, exact minimum-lift descent gives

$$K_O^{\mathrm{quot}} = K_{HH} - C K_{VV}^{-1} C^\dagger.$$

If K_{std} is frozen on the same support, units, boundary conditions and observer chart, the scoreable source excess is

$$\boxed{E_K = (K_{HH} - K_{\mathrm{std}}) - C K_{VV}^{-1} C^\dagger.} \tag{127}$$

This distinction is necessary. The positive auxiliary square

$$\frac{1}{2}\langle h, K_{\mathrm{std}} h\rangle + \frac{1}{2}\|v - Fh\|_W^2$$

has $C = -F^\dagger W \neq 0$ for $F \neq 0$, yet its Schur quotient is exactly K_{std}. Mixed parent coupling therefore does not entail a Tau-over-standard signal.

Let the regular coframe push Eq. (127) to E_g, and let $\mu_\tau = \bigoplus_a \mathcal{T}_a[E_g]$ stack all predeclared terminal images. The direct sum means stacking, not statistical independence. Exact linear duplicates are quotiented and the full cross-terminal covariance Σ is retained. After whitening and projecting a frozen nuisance design, write the held-out standard residual and Tau prediction as r and m. For the two fixed hypotheses the exact Gaussian score is

$$\boxed{S_{\mathrm{G4D}} = \chi_0^2 - \chi_\tau^2 = 2\,\mathrm{Re}\langle r, m\rangle - \|m\|^2.} \tag{128}$$

No Tau amplitude, radial profile or terminal weight is fitted. This theorem closes the conditional source-to-score compiler; it does not show that the physical base–seed parent occupies a nonzero packet or authorize a current Nature score.

Common quantum–gravity transport and gravity normalization

Suppose the occupied quantum and gravity terminal images are faithful realizations W_Q, W_G of one positive source generator K_*. Their relative map is then not an independent terminal gain:

$$T_{G\leftarrow Q} = W_G W_Q^{-1}, \qquad T_{G\leftarrow Q} K_Q = K_G T_{G\leftarrow Q}. \tag{129}$$

When both terminal metrics pull back to K_*, this map obeys $T_{G\leftarrow Q}^\dagger h_G T_{G\leftarrow Q} = h_Q$. In the declared faithful SHR–PSAL–QPHASE completion the same source unit is calibrated by the quantum phase and gravity normalizations:

$$\mathcal{A}_* = \frac{S_0}{2\pi} = \hbar = C_G \ell_H^4 = \frac{\ell_H^2}{6\kappa}, \qquad \ell_H^2 = 48\pi \frac{\hbar G}{c^3}. \tag{130}$$

This is a cross-terminal consistency theorem inside the completion. It does not derive G or $\hbar$ from the narrow parent, and it does not establish physical occupation of the maps. Separate terminal posterior marginals also cannot identify their joint latent covariance; common-event data or one explicit shared latent likelihood is required before a joint score is formed.

The remaining internal fanout freedom is conditionally removed on an occupied absolutely irreducible source sector with multiplicity-one quantum and gravity copies. Source equivariance makes each nonzero terminal map scalar-unique; cyclic no-bypass occupation removes the zero-map branch, the common pullback metric fixes the magnitude, and a source-owned oriented anchor fixes the sign. Thus Eq. (129) is the unique source-natural isometry inside that enriched packet. Repeated-copy and unfixed-anchor controls prevent extending this conclusion beyond its stated premises. The physical base–seed law has not yet been shown to occupy the packet.

The complete positive response Hessian and mixed source jet can, in turn, be represented by one endpoint-blind Schur-neutral relation-defect square whose post-body minimum leaves the seed–body action unchanged. This closes the scalar action-form search, but not gravity-role typing: rotated terminal projectors preserve the same scalar action while changing the named terminal responses. The minimal sufficient object in this declared class is therefore a complete graded source characteristic containing its source-owned role representation. Physical ownership of that characteristic remains open.

The pointed exterior/CAR result compresses the fanout assumptions without removing that boundary. Its onto no-bypass orbit is cyclic, absolutely irreducible and spectator-free. A source-owned oriented two-role plane would therefore supply one Q and one G copy, and the common

product metric would close the normalized Schur transport. Without the typed role incidence, $U(2)$ role rotations preserve the untyped source action and metric while changing the named projectors; zero-leg and duplicate-copy controls remain. The unresolved parent input is consequently one nonzero oriented Q/G incidence inside the complete graded representation, not a terminal gain. Inside the declared enriched P3 plus Tau-local composition class, the two ordered nonzero cotangent ports already realize this role plane and the local tensor law supplies one source copy per role. A hidden-diagonal control with full Q/G marginals but deficient joint rank proves that factorization cannot be inferred from marginal occupation. Thus the unrestricted physical blocker is joint realization of the known P3 port packet and locality law, not another gravity–quantum connector. The already constructed once-counted component action $\widehat{\Gamma}_{\mathrm{CLP3}} = \sum_C \widehat{\Gamma}_C$ realizes the P3 port and locality premises jointly at its attainable global zero. Consequently the pointed-CAR and Schur results close the conditional isometric Q/G transport, and the faithful phase specialization fixes $\mathcal{A}_* = \hbar$, with no new gravity–quantum gain. Equal-reduct hidden-global and missing-port controls show that the narrow parent does not select this completion. The physically identifiable target is the associated completion class rather than one off-shell formula. Adding $\lambda\Gamma^2_{\mathrm{CLP3}}$ preserves the occupied minimum, physical Hessian and typed fanout packet while changing the off-shell action. Nature-level selection therefore asks whether the parent realizes this class; hidden-global and missing-port controls remain outside the class.

Finite spectral hierarchy: a non-identified mass bridge

An independently proposed finite-complex/boundary-mode packet gives a useful comparison for the matter terminal but supplies no new terminal law here. For $Q_f = \partial_x + m$ on $[0, L]$ with the stated self-adjoint Robin domain, the endpoint resolvent has a zero-mode residue $m/\sinh(mL) \sim 2m\exp(-mL)$. Its dimensions are inverse length, and positive-spectrum pairing of $Q_f^\dagger Q_f$ and $Q_f Q_f^\dagger$ does not fix mL, the boundary traces, a dimensionless proton ratio, or a mass normalization. A different domain or terminal map can change the residue. This construction is therefore a conditional hierarchy mechanism candidate, not an additional derivation of the observed matter spectrum.

The post-body relation-row stiffness reported in the companion source is also not, by its name or dimensional appearance, the Einstein–Hilbert coefficient in Eq. (130). Equating them requires an independently typed, normalized physical handoff and compatible action, length and mass units. The present common-action identity is conditional inside its declared completion; a measured G or proton mass may be a diagnostic target, not a backward selector of Q_f or the parent packet. The corrected universe-seed evolution working packet is cited only at this comparator level. An offspring/inheritance law and its hypothetical fitness calculations are outside VII-A's one-universe terminal assembly.

The same-unit comparison is now explicit. Equation (130) and $U_R^2 = 2\ell_H^2$ imply $S_0/\hbar = 2\pi$ and

$$\kappa_{\mathrm{row}} = \frac{\hbar}{2\ell_H^2} = \frac{c^3}{96\pi G}, \qquad C_{\mathrm{EH}} = 6\kappa_{\mathrm{row}}.$$

Thus the source proposal $S_0 = \mathcal{A}_* = \hbar$ conflates two normalizations, and its direct row/EH identification cannot hold in this same completion. The compatible algebra is $G = c^3 U_R^2/(48 S_0)$; none of these identities selects a measured constant.

A finite conditional matter witness goes one step beyond the bare residue. On a separate post-body scalar field, a declared positive boundary term gives $H_g = H_- + gww^*$, where w is the fixed endpoint evaluation covector's adjoint. Writing $a = \langle u_0, w\rangle$ and $b(\lambda) = \langle w_\perp, (H_\perp - \lambda I)^{-1} w_\perp\rangle$,

exact Feshbach reduction yields

$$\lambda_g = \frac{g|a|^2}{1 + gb(\lambda_g)}.$$

For $0 < g|a|^2$ below the positive gap this has a unique positive subgap root. At constant $W = m$ and large mL, $\sqrt{\lambda_g} \sim \sqrt{2gm/(1 + g/(2m))}\exp(-mL)$. The frozen finite-Q benchmark reproduces this conditional mechanism and the original low-spectrum convergence. The extra coupling, scalar kinetic law, particle identity and Nature occupation remain assumed; this is not a proton mass prediction.

Unified-selection and pair-test handoff

PD-USS1 conditionally places quantum, phase and Parent-Direction exits on one source-selected transport ancestor, but it does not prove that two named terminal outputs carry independent source information. For the P3 cross-role test the stacked typed rank must therefore be established before the terminal product is scored. Technical Paper VII-B derives the weaker product-free disk bound and updates the targeted public census to 0/31 eligible packets. In particular, two optical homodyne arms or a mechanical state decoded through a qubit remain common-carrier precedents rather than independently typed gravity/matter/quantum occupation certificates. This changes no joint-terminal assembly or standard-limit result in VII-A.

ACT-PCLOSE1 terminal-stack corollary

The upstream ordered action theorem fixes the common regular response by

$$D_S Z_* = -H_Z^{-1} B_{ZS}.$$

Every time, quantum, gravity, matter, gauge and hybrid terminal in the present assembly is therefore a source-frozen typed pushforward $O_i = Q_{OS,i} \circ T_i \circ P_i \circ \Xi(Z_*)$. The exact local completeness criterion for the assembled family is

$$\boxed{\ker J_{\mathrm{term}} = \mathcal{N}_{\mathrm{phys}}.}$$

The equality has not been proved for Nature's full terminal atlas. Even if it holds on the declared chart, it does not establish global nonlinear completeness, ambient source exhaustivity, or a nonzero standard-orthogonal signal.

17 Two graph Laplacians and the spacetime boundary

For a conditional normalized Gram packet write $G_{ab} = w_{ab}U_{ab}$, where $w_{ab} = |G_{ab}|$ and U_{ab} is a unit phase on nonzero support. Two operators with different roles must be separated:

$$L_{\mathrm{geom}} = D - W, \qquad L_U = D - W \odot U, \qquad D_{aa} = \sum_b w_{ab}. \tag{131}$$

The real operator satisfies $L_{\mathrm{geom}}\mathbf{1} = 0$ and can generate a probability-conserving diffusion. The magnetic operator is Hermitian and positive because

$$\psi^\dagger L_U \psi = \frac{1}{2}\sum_{a,b} w_{ab}|\psi_a - U_{ab}\psi_b|^2 \geq 0, \tag{132}$$

but generically $L_U\mathbf{1} \neq 0$ and it is not the same diffusion generator. Using one symbol for both would mix geometric scaling with phase-sensitive frustration.

A heat-kernel diagnostic may be defined from the real diffusion operator,

$$P(s) = N^{-1} \operatorname{Tr} e^{-sL_{\mathrm{geom}}}, \qquad d_s(s) = -2\frac{d\log P(s)}{d\log s}. \tag{133}$$

A robust plateau $d_s \simeq 4$ would establish only an effective four-dimensional diffusion regime. It does not distinguish Euclidean four geometry from Lorentzian $3+1$ spacetime and supplies no causal covector, light cones, time orientation, global tetrad, spin lift or Einstein dynamics. Those require the independent DA, G4a/G4b and topology gates inherited by this paper. Similarly, nonzero U-holonomy is a projective graph invariant, not yet terminal gravitational curvature.

Thus the Graph/Hilbert construction can provide useful common diagnostics for quantum, phase and geometric terminals, but it does not merge their physical meanings by notation. A hybrid terminal still requires one source-owned shared descriptor, typed projections and an occupied mixed response block. Neither a numerical spectral plateau nor a loop phase is a Nature-level quantum–gravity prediction without that forward source chain.

18 Common Q/coframe ancestry and compatible cosmological atlas

The ECTR specialization makes one part of the joint-terminal ancestry exact inside a declared source completion. Let the frozen body supply

$$A_{0,M} = J_{M,S}^{*} h_S J_{M,S} > 0, \qquad q_{0,M} = A_{0,M}^{-1}, \tag{134}$$

and let a normalized post-body Legendre–Riesz coevaluation give

$$\boxed{G = \frac{K_{\mathrm{src}}}{2A_*} = k_{c,O}^2 A_{0,M}^{-1}.} \tag{135}$$

The metric and Q terminals then remain differently typed descendants,

$$e_{a,M} = a e_{0,M}, \quad e_{0,M}^{*} e_{0,M} = q_{0,M}, \qquad F_{Q,O}(a) = (I + a^2 G)^{-1/2}. \tag{136}$$

This supplies a common source spectrum without identifying quantum response with gravity or adding a terminal-specific gain. Quantum effects still need the phase/effect/instrument packet of Paper VI; gravitational dynamics still need the metric action and stress coupling declared in VII-A.

The source diagram is therefore a common-cause diagram: the metric expansion readout and the expansion-conditioned terminal weights descend in parallel. The scale a_O modulates their relative dominance within the frozen law; it does not select the source operator, quantizer or terminal semantics and does not turn one operational theory into the other.

If the occupied rank-four leaf is connected, oriented, time-oriented and globally hyperbolic, choose a comoving coframe

$$e_{\cos}(u) = (n^{\flat}, a(u)\bar{e}^1, a(u)\bar{e}^2, a(u)\bar{e}^3). \tag{137}$$

On overlaps, compatible observers have $e_j = \operatorname{diag}(1, R_{ij}) e_i$ with $R_{ij} \in SO(3)$ and

$$A_{0,j} = R_{ij} A_{0,i} R_{ij}^{-1}, \quad G_j = R_{ij} G_i R_{ij}^{-1}, \quad F_{Q,j} = R_{ij} F_{Q,i} R_{ij}^{-1}. \tag{138}$$

Therefore spectrum, determinant and the sign of the loss-over-elasticity expansion classifier agree across that comoving atlas. Arbitrary bound, accelerated or tilted observers are not covered.

The result is conditional and kinematic at the expansion-sign level. The same body and coframe permit a different source coevaluation, and the same operator packet permits zero access. The present physical source does not yet select the TLKR development, coevaluation, positive occupied access and global atlas as one Nature packet. No Einstein equation, accelerated expansion, dark-energy density or concrete $H(z)$ follows from this section.

19 Observer realization: inherited results and physical boundary

The current observer programme distinguishes a conditional mathematical construction from an occupied physical measuring system. The source order remains base plus one universal seed, stabilized morphological response, observer–source restriction, shared descriptor, and resolved terminal readouts. Neither a detector nor an empirical residual constructs the parent body backwards. Dynamics below concern downstream subsystems only.

The canonical observer-origin derivation, Sections 22–26, owns the following results. The existing joint source action fixes static and frequency-dependent response together. Variation of the BRAC potential couples to the field square; a linear field coupling is a distinct comparator unless derived by expansion about an occupied background. Fixed canonical means select a unique coherent state in a finite positive harmonic packet only if there is no excess fluctuation energy. This is a free/asymptotic preparation statement, not the ground state of an interacting detector. For the existing coupled harmonic pair, the joint ground is pure but either local subsystem is mixed, with its covariance fixed by the complete joint Hamiltonian.

These are conditional derivations and internal numerical checks. The actual observer–source context, physical preparation process, kinetic/energy calibration and stable operational resolution remain unproved. The existing ORA-T1 selector conditionally chooses stable quantizers when the occupied context, admissible partition family and positive closure barrier are supplied; it does not establish those physical premises. A computed ground state is not a preparation mechanism, and a covariance or optimal abstract measurement is not an implemented stable quantizer Q_{OS}.

Ownership and use. The central observer-origin note owns the new contact, state and covariance derivations; the body-readout technical companion carries the consolidated mathematics. Foundation Papers IV, V and VI retain physical-selection, clock-descent and quantum-readout ownership, respectively. This paper imports only the consequences within its scope. No observational dataset, baseline, frozen coefficient, score or empirical conclusion is revised by this update. The reproducibility supplement records the source hashes and the exact conditional/physical distinction.

Consequence for this paper. These results do not select a particle spectrum, coupling constant, Higgs vacuum or gravitational terminal. Common source ancestry remains weaker than physical realization of those sectors.

20 Finite source-to-readout implementation and its scope

The current finite laboratory is a reproducible implementation, not a new Nature-selection theorem. The common-cone example uses the full frozen body Hessian but supplies the local support

and kinetic/gradient forms. A same-body kinetic rescaling changes its chart speed. Maxwell polarization, quantized photon energy and contact exchange therefore remain conditional recoveries rather than source-selected SI constants, gravity or detector occupation.

Ownership and reproducibility. The seed–body companion owns the finite minimum/Hessian example, the body–readout companion its ordered access and contact reduction, and the terminal companion its conditional propagation boundary. The physical-source papers retain the source-selection obligation. Code, declared priors, adverse controls and provenance are available at `https://github.com/tau-core-research/tau-core-theory/tree/main/experiments/seed_evolution_lab`. Numerical tests are internal evidence only; no published empirical score is changed.

21 Source identification and recovery

The current observer audit separates a conditional forward construction from physical source selection. A frozen body, occupied observer incidence, preparation, complete instrument and stable resolution can determine records without selecting those inputs from Nature. Equal effects do not fix postmeasurement states, and symmetry within a chosen interaction family does not exclude other symmetric couplings. These results do not select a particle spectrum, coupling constant, Higgs vacuum or gravitational terminal. Common source ancestry remains weaker than physical realization of those sectors.

Dependency and claim ownership. Foundation VI and the body–readout technical companion own the finite instrument/access proof; Foundation IV and TD-0 retain physical source selection; Foundation V, the temporal synthesis and seed–body companion own the gradient/inertial distinction. This paper inherits only the stated specialization. Canonical assumptions, countermodels and calibration audits are in `https://github.com/tau-core-research/tau-core-theory/blob/main/docs/tau_rqm_instrument_to_record_bridge_001.md`. No physical seed, Nature occupation or Tau-exclusive observation is established by this update. Existing empirical endpoint definitions and scores are unchanged.

Imported result (CPR1). Paper 4, *Full Derivation Ledger for the Branch A Higgs Module*, section "Conditional source preparation and time-resolved access, Theorem CPR1(a,c), v1.0 (2026-09-28), is the sole proof owner. The common scalar–Dirac operator admits conditional pair transfer and time-resolved emission. It does not thereby select particle occupation, couplings or a vacuum; inclusive widths and coherent mode coupling remain distinct inputs. No additional particle or gravity recovery is claimed. Physical preparation, detector/clock calibration and stable Q_{OS} remain assumptions; no Tau-exclusive observation follows.

Current source/readout status (9 October 2026). Foundation VI, OIR1(a–c), v1.0 ("Output statistics, surviving-source instruments and apparatus scope") and Paper 4 full derivation, MSI1, v1.0 ("Source work, bilinear mass pairing and independent selection"), both dated 9 October 2026, own the proofs. The common source architecture does not identify the optical count or QND pointer as a gravitational terminal. MSI1 keeps mass, kinetic form and current jointly typed; OIR1 keeps the complete instrument distinct from output effects. A physical metric/stress completion, operational units and Nature occupation remain open. These are conditional mathematics

and standard-compatible controls, not Nature occupation or Tau-exclusive evidence. Stable Q_{OS} and physical calibration remain open; empirical scores are unchanged.

22 Conditional finite source–state result chain

The 10 November 2026 ET/CAR record supplies a bounded downstream example, not a parent-source selection or empirical cosmological result. Foundation VII-A owns the detailed response presentation; its HSL section and compact receipt retain the numerical provenance. The inherited stages have different logical status:

1. HRZ/HSA derive the bosonic Hessian and the once-summed metric/scalar mean constraints from the same source functional. The original zero-derivative reference is not a saddle; positive volume alone does not repair it. This does not exclude the HORIZ preparation family.

2. HSD/HSE solve regular initial jets for that original preparation at $k = 16, 32$ and check all three mean Euler equations and Hamilton-constraint propagation. At 90-digit precision the checked residuals are below 4×10^{-85}. A frozen-state derivative control fails.

3. HSF evolves the background and occupied frame together on $0 \le t \le 0.02$, without instantaneous state refill or projection. RK4 refinement gives a ratio 15.979; the finest lapse residual is 4.29×10^{-10}. Freezing the state instead gives a residual about 4.58. HSG independently replays every fine-grid step, with discrepancy below 1.4×10^{-17}. The original numerical-backend warnings remain qualified, not explained away. HSH tests common-constraint regularity at the saved nodes, not a rigorous between-node or global stability bound.

4. HSI/HSJ retain nonzero common-source covariance and mandatory cross terms in the same transported Slater state. A vanishing mean constraint does not imply vanishing quantum noise. HSI's coarse-grid mean-binding check failed its unchanged tolerance; the separate finest-grid guard passes without erasing that failure.

5. HSL transports the full normal16 carrier on this same background and evaluates all nine retarded source blocks at three fixed times. Eight internal checks pass, with refinement ratio 16.4278 and state-transport discrepancy below 10^{-15}. This is not a continuum error certificate. The response formula retains the original preparation bracket, upper endpoint and local contacts; numerical contact/pulse assembly and the common metric Volterra solution remain open.

The assumptions are a finite homogeneous normal Hamiltonian, the original HORIZ state prescription, one source functional and original endpoint maps. An alternative admissible state can change covariance and retarded expectation. Neither normal finite evolution nor off-diagonal response establishes physical particle creation. Spatial and renormalized quantum–gravitational closure, physical state and Q_{OS} selection, general/all-state/UV stability, GR/UV completion and cosmological validation remain open.

The following unequal-time response section uses precisely this same path and preparation. The finite initial consistency and trajectory checks do not solve the full metric Volterra response.

23 Finite homogeneous unequal-time source response

This section records the conditional downstream ET/CAR calculation HSL (10 November 2026), not a derivation of its matter action or physical initial state from the Tau parent. The par-

ent remains atemporal. We use the same normal 16-component Hamiltonian, original HORIZ preparation family and rank-eight Slater projector as the saved short coupled background HSF; normal evolution is not an anomalous pair-creation Hamiltonian. In source units the fixed inputs are $N = B = M_P = V_{\rm com} = 1$, $m = B_\phi = m_\phi = 0$, $k = 16$, $a(0) = p(0) = 1$, and $h(0) = 0.6988328400563818$. Here $h = \dot a/a$ and $p = \dot\phi$. The background solves the coupled mean Euler equations on $0 \le t \le 0.02$; it is not a fitted cosmological history.

Original source and endpoint ownership. Let $\dot\rho = [G, \rho]$, $G^\dagger = -G$, and let $u_b = (\delta N, \delta \log a, \delta\phi)_b$. Write the original source variation as

$$\delta G = \sum_b \sum_{j=0}^{2} G_{bj} \partial_t^j u_b.$$

With $\mathcal{L} = \partial_t - \mathrm{ad}_G$, the independently derived Euler matrices E_b obey

$$G_{b0} - \mathcal{L} G_{b1} + \mathcal{L}^2 G_{b2} = -iE_b.$$

This identity does not define a new contact from a desired Ward equation. Set $X_b = G_{b1} - \mathcal{L} G_{b2}$, $Y_b = G_{b2}$ and

$$\eta = \zeta + \sum_b \big([X_b, \rho] u_b + [Y_b, \rho]\dot u_b\big), \qquad \dot\zeta = [G, \zeta] - i \sum_b [E_b, \rho] u_b.$$

If $\partial_t U(t,s) = G(t)U(t,s)$, $U(s,s) = I$, then the retarded bulk kernel is

$$K_{ab}(t,s) = -i\theta(t-s) \operatorname{Tr} \rho_i [E_a^I(t), E_b^I(s)], \tag{139}$$

$$E_a^I(t) = U(t,i)^\dagger E_a(t) U(t,i). \tag{140}$$

The complete first expectation variation retains, separately,

$$\begin{aligned} \delta\langle E_a(t)\rangle = {} & \operatorname{Tr}\Big(E_a(t) U(t,i) \zeta_i U(t,i)^\dagger\Big) + \sum_b \int_i^t K_{ab}(t,s) u_b(s)\, ds \\ & + \operatorname{Tr} \rho(t) \delta E_a(t) + \sum_b \operatorname{Tr} \rho(t) \big([E_a, X_b] u_b + [E_a, Y_b]\dot u_b\big), \end{aligned} \tag{141}$$

$$\zeta_i = \delta\rho_i - \sum_b \big([X_b(i), \rho_i] u_b(i) + [Y_b(i), \rho_i]\dot u_b(i)\big). \tag{142}$$

Thus the inherited preparation is not silently replaced by $\zeta_i = 0$. For lapse variation $Y_N = 0$ and $X_N = G_N$ (the coefficient of $\delta\dot N$); the original upper endpoint and lower sewing bracket survive. For the common constraints $F_a = V_{\rm com} E_a^{\rm bos} - \operatorname{Tr} \rho E_a$, the fermionic bulk contribution is $-K$, and the bosonic Hessian and original local contacts enter once. No second measure, Schur term or independent noise leg is added. In natural units $[E_N] = [E_{\log a}] =$ mass and $[E_\phi] = 1$, while $[u_N] = [u_{\log a}] = 1$, $[u_\phi] =$ mass; hence $[\int ds\, K_{ab} u_b] = [E_a]$. Commuting transported insertions give zero bulk response, not necessarily zero contact or quantum noise.

Finite numerical evidence and limits. HSL reconstructs the original saved RK4 stages at 20, 40 and 80 steps, without extending the background, refilling the state or unitarizing U. All nine source blocks are evaluated at $t, s \in \{0, 0.01, 0.02\}$; the propagator is saved at every grid node. Eight internal checks pass. The maximum unitarity and transported-state discrepancies are respectively

3.872×10^{-11} and 9.994×10^{-16}. Direct density-commutator and interaction-picture kernels agree to relative 8.562×10^{-12}. Successive absolute kernel differences are 0.19284046 and 0.01173866 against a maximum kernel magnitude 7.08546827×10^{8}, a refinement ratio 16.4278 consistent with fourth-order convergence. Omitting transport changes a tested kernel by 6.59940907×10^{8}: equal-time covariance is not a substitute for this response. Causality is imposed by the retarded definition; it is not independently proved by the zeroed future triangle. The diagonal convention $\theta(0)=1$ adds no delta contact.

The HSF numerical-backend warning history remains qualified; an independent contraction replay confirmed its finite saved arrays and every fine-grid step, but did not establish the cause of the original warnings. The archived HSI equal-time audit also retains its failed coarse-grid mean binding check; the separate fine-grid guard does not erase that failure. The compact HSL receipt is included as provenance, not as a portable replay of the entire private source chain. Numerical contact/preparation assembly, a direct compact-pulse response check and the solved common metric Volterra response remain open. No spatially complete or renormalized quantum–gravitational closure, physical-state selection, particle production, general stability, GR/UV completion or cosmological validation follows. Changing the admissible initial state can change the expectation kernel.

24 Conclusion

Paper VII-A establishes a conditional common-source architecture for gravity, matter, Abelian gauge response, formal clock response and quantum backreaction, together with the exact bookkeeping and source-rank conditions needed to avoid double counting. Its strongest positive bridge is the ROOT–TT same-preparation theorem; its strongest negative result is that standard-limit agreement and finite terminal closure do not identify common parent provenance. Within the inherited seed-fixed Hodge-generated class, Clifford naturality also removes an independent geometric root and enforces a common photon/observer cone; this does not close physical class membership. The same conclusion holds for measure provenance: the inherited CCOC–OGAS section fixes local orientation, and conditional measure matching fixes its common-line Jacobian, but Paper VII-A neither constructs nor physically occupies the full projective/body connector.

The original monolithic Paper VII has therefore been separated without discarding information. The P3 algebra and law-to-packet proof now belongs to VII-B, and the scalar-15/RG branch to VII-C. The decisive remaining result is still either a deeper physical base–seed derivation of the jointly occupied source packet or a source-frozen, gain-free cross-terminal prediction not implied by standard physics.

The temporal refinement leaves the assembly order intact. A temporalizable completion may add EOCC actuality upstream of the observer clock, but the joint Einstein–Maxwell–matter terminal neither derives that actuality nor turns its formation parameter into stress–energy. Any dark-sector use would require an independently frozen covariant source current whose conservation, units and standard-limit subtraction are proved within this same once-counted ledger.

A Self-contained dependency certificate

This appendix separates logical premises from provenance. The conclusions of Paper VII use only the following statements. A reader may treat each imported statement as an explicit assumption and verify every theorem in this paper without consulting another Tau manuscript.

D1: body-first order.
There is a C^2 pre-body action $\mathcal{A}_{\mathrm{pre}}[y; B, s]$ with an occupied stationary point y_* and positive physical Hessian H_B. The post-body functional is evaluated on the frozen parameter y_*; post-body source variation is not varied back into the body equation. This is the only body-first premise used in Eqs. (16) and (22).

D2: common typed source domain.
On the selected body and observer support there is one finite source domain $\mathcal{E}_{*,O}$, one access map A_O, typed linear role maps $R_a : \mathcal{E}_{*,O} \to \mathcal{Y}_a$, and stable resolution maps $Q_O^{(a)}$. No direct-sum decomposition of $\mathcal{E}_{*,O}$ is assumed. Proposition 1 is proved here from exactly this premise.

D3: occupied operation data.
An operation algebra $\mathfrak{A}_{\mathrm{src}}$, occupied vector Ω_s and positive root covariance X_0 are given. Whenever the full-occupation branch is used, Ω_s is cyclic and every required R_a is surjective. Whenever the rooted-path branch is used, $\operatorname{supp} X_0 \subseteq V_s$ and every admitted L_e preserves $V_s = \operatorname{span}(\mathfrak{A}_{\mathrm{src}}\Omega_s)$. Proposition 2 proves the resulting full-occupation or activation statement; physical realization of either premise set is not imported.

D4: fixed quantum preparation.
The CTP boundary datum satisfies $\rho_{\mathrm{in,O}} \geq 0$ and $\operatorname{Tr} \rho_{\mathrm{in,O}} = 1$. It is held fixed while g, A and Φ are varied. The CTP evolution and observer restriction produce a normalized output state ρ_O. CTP response is obtained by varying the difference source before taking the physical equal-branch limit. No Euler variation with respect to a density matrix is used.

D5: standard Lorentzian leaf.
On the selected leaf the geometric and Abelian functionals satisfy the classification hypotheses of Proposition 5, with a common unit packet. Their Einstein–Hilbert and Maxwell shapes then follow inside that class. The matter functional, measure and renormalization prescription are diffeomorphism and $U(1)$ gauge invariant and anomaly-free for the Ward identities invoked here. This is an assumed standard-limit leaf, not a Tau derivation of those actions.

D6: operational effect source.
Each declared outcome family is a POVM, $E_a \geq 0$ and $\sum_a E_a = I$, and its source coupling is Eq. (17). This premise, together with D4, is sufficient for the normalized Born derivative in Eq. (20); no stronger quantum reconstruction theorem is used.

D7: clock-source boundary.
The clock character ϑ is an admitted external source coordinate of the post-body functional. Paper VII uses only the formal response derivative $\delta W_O^{\mathrm{ren}}/\delta\vartheta$. It neither imports nor proves a duration law, global clock integrability or a Tau-specific clock distortion.

D8: once-counted calibration.
Every displayed sector uses one declared action unit, each occupied contribution and counterterm has one ledger owner, and the overlap $K_{m,m+2}$ is calculated from Eq. (58) and frozen before the ROOT–TT comparison. This bookkeeping premise forbids an additional fitted gain but does not select the common unit, occupied mode pair, screen or subtraction prescription.

Provenance boundary

The eight D-items above are fully stated premises. A reader may verify the conditional mathematics without consulting another Tau manuscript. Their provenance does not promote them into unrestricted physical parent laws.

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